100 problems.
A stronger second draft.

Prepared for Fred Esser | Revision 2 of the first working round

The original simulated authors returned to the review, revised every problem card, and used the criticism to run sharper tests. This edition replaces the earlier notebook while preserving its research ambition and its explicit limits.

100 revised cards	24 review responses	3 new investigations
Exact scope, restricted question, baseline, useful outcome and missing bridge	Changes and remaining gaps are recorded against every reviewer finding	Minimum sampler size; failed product composition; elliptic-curve control

What changed in the work

The corrected three-bit sampler problem now has an exact answer: four equally likely selected strings are necessary and sufficient for error at most 1/4 in the stated model. The follow-on product sampler fails to keep that error at six bits: its exact worst discrepancy is3/8. The team has retained both outcomes and changed the next question.

The original three pilots reproduced their full JSON outputs. Twenty-five named acceptance/rejection controls passed. The new sampler result has a structurally different verifier, and a separate AI pass checked the composition calculation.

How to read the revised catalog

A route marked protocol design still needs the named inputs, definitions or mechanism before execution. Calibration ready means a bounded known-example check is specified. Research protocol ready means a bounded proposal is specified; feasibility, novelty and success remain unestablished. A separate completed-result field identifies actual work.

All participants and the address are simulated. These are AI passes within the same workflow. No major open conjecture is claimed solved, no new contribution is claimed established, and no guarantee of solving all 100 problems is given. Finite evidence remains finite.

The team returns to work

Fictional motivational address for the simulated original team.

Imagine the sting of having a promising plan returned covered in corrections. In this scenario, the team has a choice: defend the first draft, or make the work stronger. We choose the stronger work.

Our ambition remains to solve these problems. That ambition gives us reason to sharpen every definition, expose every missing bridge, and ask questions that survive contact with evidence. A criticism that identifies a hidden assumption has given us a more precise target.

Bring unconventional connections. Defend them with clear arguments. Help the critic find the weakest step, then build something better where it broke. Keep the difficult questions alive without pretending a small experiment has answered them.

This round, we will separate completed results from planned tests and concentrate effort where an answer can change our next decision. The horizon remains enormous. The next deliverable must be exact. Let the boldness of the mission and the discipline of the work strengthen each other.

The working commitment

Treat criticism as a source of specific tasks. Preserve ambition, unconventional ideas and dissent. Give credit when a correction improves a definition or an experiment. A failed approach is useful when it identifies a precise obstruction and changes what we do next.

The actual process behind the roles

Stage	Executed passes and visibility
Original draft	Six author passes across subject areas and methodology implemented ten conceptual research roles.
Peer review	Six reviewer passes; the four technical reviewers read different subject sections from their author assignments. The reproducibility reviewer had authored the Collatz audit.
This revision	Five original domain-author passes revised their sections; one methods pass added controls and reviewed new results; the integrating assistant executed the sampler and composition tests.
Independence limit	Review and revision passes could read the supplied earlier material and findings. The ten roles do not imply ten independent people or ten separate proofs for each problem.

Method, readiness and next decisions

1. Maintain exact target, restricted question, baseline, informative outcome, unresolved bridge, and execution status for every catalog item. Separate result status from planned-route status.

2. Activate one principal project and one inexpensive control per bounded round; keep other cards as scoped proposals until explicitly selected.

3. Before an activated computation, freeze model, inputs, arithmetic, outputs, resource budget, and falsification criteria. Before a proof attempt, freeze hypotheses, desired lemma, dependencies, and a counterexample search.

4. Generate alternative ideas before sharing the chosen argument when independent drafting is actually performed; record the drafts and visibility boundaries. Ten conceptual roles do not imply ten actual independent passes.

5. Require a critical pass to test model restrictions, target changes, circular dependencies, and evidence labels. Record the response, what changed, and what remains unresolved.

6. Run exact finite controls before interpreting the main result. Save code, canonical outputs, execution environment, and verifier differences. A run cut off by its cap must be labeled incomplete.

7. Advance only if the result changes a stated decision: it certifies a bounded fact, falsifies a proposed mechanism, identifies a missing lemma, or materially narrows an obstacle. Otherwise park or reformulate at the round boundary.

8. Complete a targeted closest-prior-work audit before describing any result as new. A solved benchmark may train or validate the workflow without becoming a research advance.

9. Every report distinguishes proposed, executed, finite verified, conditional, and proved statements; it names the bridge still required to reach the unrestricted target.

Current catalog status	Cards
Protocol design	65
Calibration ready; unexecuted route	28
Research protocol ready; unexecuted route	7

Five cards contain separately recorded completed results. This count includes retained and newly executed work; it does not imply that the proposed next routes have run.

New result / minimum sampler support

Problem 81: a corrected, fully specified finite calibration

Read three input bits x1,x2,x3 once in that order. A deterministic branching program has two states at every layer, starts in state 0, chooses any transition for each current-state/input pair, and accepts any subset of final states. Compare uniform input on all eight strings with uniform input on a selected nonempty subset S. Minimize |S| subject to acceptance error at most 1/4 for every such program.

Each layer has 16 possible transition maps. Enumerating 16^3×4=16,384 descriptions yields 88 distinct Boolean functions. Every one of the 255 nonempty supports was checked using exact rational arithmetic.

Uniform support size	Best worst-case error	Number attaining that best error
1	7/8	8
2	1/2	12
3	1/2	48
4	1/4	24
5	3/10	48
6	1/6	12
7	1/8	8
8	0	1

Minimum support is4. Exactly 24 four-string sets achieve the 1/4 threshold. One example is S={000,001,010,101}. Every support of size 1,2 or3 fails; the package records 92 explicit violating-function witnesses. Two fair seed bits can select uniformly among the four strings in this one fixed model.

Independent characterization and verification

After two input bits, the four prefixes induce one-bit residual truth tables. Width two permits at most two distinct residual tables. This is also sufficient: the first layer distinguishes the first bit, the second chooses one of the two residual functions, and the final layer evaluates it. There are 4 constant choices plus C(4,2)×(2^4-2)=84 uses of exactly two residual tables, giving 88 functions. The verifier uses this characterization and explicit rational means rather than the first program enumeration.

The nonmonotone entry at support 5 is possible because every selected point must have equal mass. Adding a point changes all masses; larger uniform support need not improve the optimum at every consecutive size. This result is a finite calibration with no novelty claim.

New result / composition fails at fixed error

The next question was tested immediately

Does independently using S={000,001,010,101} on each of two consecutive three-bit blocks retain the 1/4 error guarantee? The sampler uses the 16 equally likely strings S×S. The target programs now read six bits in order, still with two states per layer.

Every six-layer program splits into two three-layer blocks. For each block retain transition probabilities from both initial states under uniform eight-string input and under uniform S. All 4,096 transition descriptions collapse to147 distinct paired kernels. Checking all 147^2=21,609 ordered kernel pairs exactly gives worst acceptance discrepancy 3/8.

Explicit counterexample	Value
First three transition codes	0, 1, 2
Last three transition codes	1, 3, 3
Start and accepting state	Start 0; accept final state 0
Full-input acceptance	40/64 = 5/8
Product-sampler acceptance	4/16 = 1/4
Absolute discrepancy	3/8 > 1/4

Encoding: bit(2×state+input) of each four-bit transition code is the next state. Directly evaluating all 64 inputs and the 16 sampled inputs reproduces the displayed counts. A separate AI pass regenerated the 147 kernels and recomputed the maximum with rational two-state distributions.

What the failure changes

This particular independent product does not preserve the one-block error bound. The next proposal allows a correlated sixteen-point support across six bits, or explicitly tracks growing error. Other four-point samplers, other six-bit supports and asymptotic derandomization are not ruled out. A generic additive-error composition bound was never the rejected claim.

The stored claim is a finite counterexample to fixed-error preservation for this S, with an exhaustive maximum in its stated six-bit model. It is not a P-versus-BPP separation and carries no novelty claim.

New control / traces and a local obstruction

Problem 23: distinguish an impossible event from a small-sample zero

For E2: $y^2=x^3-2x+1$, the cubic factors as$(x-1)(x^2+x-1)$. The point(1,0) has order two. At good primes other than 2 and 5, the finite group has even order, so a_p=p+1-#E2(F_p) is even. Trace 1 is therefore impossible for this curve. It is now a named known-zero control.

Exact point counting through p<=1000 used 166 included good primes for each selected curve. On E2 every trace was even, with zero trace-one events, five trace-zero events and six trace-two events. On E1: $y^2=x^3-x+1$,54 traces were odd, yet its observed trace-one count was also zero in this range.

That last zero is an observation, not the same parity obstruction. The calibration demonstrates why local algebraic constraints must be included before interpreting rare-event counts. The script stores every included prime and its exact trace. It does not establish Lang-Trotter asymptotics.

Retained pilot / Collatz growing runs

Define the accelerated odd map T(n)=(3n+1)/2^v2(3n+1). The tempting universal claim is: one fixed k makes every odd starting value n>1 fall below its start within k accelerated odd steps. For m>=2 and 0<=j<=m-1, the following identity blocks that strategy:

$$T^j(2^m-1)=3^j\times2^(m-j)-1$$

For j<=m-2, applying 3n+1 gives 2×(3^(j+1)×2^(m-j-1)-1); the parenthesized factor is odd, so exactly one factor of two is removed. Each of these steps increases the value. Choosing m=k+1 gives k initial increases. This elementary argument does not establish eventual behavior or novelty.

The retained audit checked 130,816 exact transitions for m=2,...,512 and 50,000 odd starts 3,...,100001. Starts not yet below their initial value after 1,2,3,5,8,12 accelerated steps were respectively 25,000;18,750;12,500;7,421;4,481;2,569. The full original JSON reproduced exactly.

Retained pilots and stronger controls

Problem 37 / union-closed families

Enumerate all families of subsets of a fixed labeled ground set; exclude the empty family and the family consisting only of the empty set. Retain exactly the families closed under pairwise union.

Ground-set size	Eligible families	Lowest largest element frequency
1	2	1/2
2	12	1/2
3	120	1/2
4	4958	1/2

At size 4 all 4,958 eligible families satisfy the half-frequency condition;51 attain equality. Families may use fewer than four ground-set elements. These are labeled counts, and no unrestricted theorem is inferred.

Problem 46 / balanced pairs in partial orders

Elements	Non-total labeled orders	Lowest best-pair balance
2	1	1/2
3	13	1/3
4	195	1/3
5	4111	1/3

Each of the 4,111 non-total labeled partial orders on five elements has an incomparable pair with ordering probability in[1/3,2/3] under uniform linear extension. Counts and exact fractions reproduced the original JSON.

Twenty-five named verification checks

The revised scripts expose their existing production predicates for testing. Hand-specified controls exercise empty and trivial families, union-closure failures, cycles and nontransitivity, total versus non-total orders, Collatz terminal 1 and boundary cases. All 25 checks passed; the core enumeration logic produced unchanged results.

The prior review also used alternative implementations for union-closed sizes 1-4, posets through size 4, and the finite Collatz table. Five-element posets have been rerun with the delivered implementation. These checks are ordinary executable verification, not proof-assistant certification.

Next bounded assignment

Design a correlated sixteen-point support against an explicit finite constraint bank

The executed product experiment exposed error accumulation. Allowing correlation changes the failed mechanism while keeping support size and six-bit model fixed; exact constraints can reject candidates before any general claim.

Model. Same six-bit width-two fixed-order program model; candidate is an arbitrary 16-element subset A of{0,1}^6 sampled uniformly, with binary membership variables z_x. Unlike S×S, A need not factor across blocks.

Deliverable. Create a solver-readable integer feasibility model: Σ_x z_x=16 and -16≤4Σ_x z_x f(x)-Σ_x f(x)≤16 for each Boolean program function f in a frozen bank. Start with all 88 one-block functions applied to either block and the executed composition witness. Publish the exact bank, one satisfying support or a checked infeasibility certificate if obtained, and direct truth-table discrepancies for every candidate.

Constraint bank. For an additional fixed bank of10,000 program encodings, set v0=1 and v_(j+1)=(1664525v_j+1013904223) mod 2^24. Decode each 24-bit value into six four-bit transition codes; code bit at index(2 state+input) gives the next state. Start at0 and accept state 0. Save duplicate-function counts. This is a reproducible limited bank, not a representative or exhaustive distribution.

Prerequisites. Select an available integer solver with documented status semantics; specify its exact model encoding and any infeasibility-certificate checker. Until this is done the assignment remains protocol_design, not an executed or guaranteed-feasible search.

Caps: at most 5 candidate supports,10,000 additional deterministic program encodings,120 seconds and 256MB. The solver and certificate checker remain to be selected, so this is a protocol-design assignment. No further run is implied by this schedule.

Decision. If a candidate violates a bank constraint, add the exact witness and repair or reject that candidate.

Decision. If verified infeasibility occurs for this bank, no uniform 16-point support can fool the entire model at error 1/4; preserve the finite certificate and consider larger support or a relaxed bound as a separately stated design choice.

Decision. If a candidate passes the entire limited bank, label it provisional; an exhaustive oracle over all program functions or a valid structural proof is still required for a universal six-bit guarantee.

Decision. If no candidate or certified infeasibility is obtained within caps, report unknown, retain the model and checked constraints, and reassess before increasing budget.

Cheap control / Parity residual-function width control

f(x1,...,x8)=x1 XOR ... XOR x8. For all eight-variable orders and every prefix length, substitute each prefix assignment and collect exact truth tables of the residual function on remaining variables.

For all 40,320 orders report maximum residual width at each cut; verify width one at the empty prefix and width two at every later cut when constant residual truth functions are counted distinctly. Save a minimal failing order and prefix if any mismatch occurs.

Proposed cap:40,320 orders,60 seconds,128MB. A mismatch triggers debugging. Success calibrates representation counting and supplies no complexity separation.

Response to every reviewer finding

Review IDs refer to Math_Frontiers_Peer_Review.pdf. All 24 findings receive a response below. An accepted correction can be implemented while the broader mathematical bridge or novelty audit remains open. Original page numbers in the review refer to the prior 59-page edition.

R1.1

Change. Added targeted prior-work audit as a gate before any originality claim; precise source selection remains an explicit protocol-design dependency where absent.

Remaining. No full novelty audit or journal contribution assessment is complete. The deliverable remains an exploratory notebook.

R1.2

Change. Added restricted questions, baselines, informative outcomes, and execution-status labels. Items 2 and 4 receive concrete calibration objects; items 1,3,5 remain protocol_design until their named source/certificate dependencies are fixed. The selected composition/control assignments specify exact models and budgets.

Remaining. Source selection or mechanism design remains explicit for unready cards. Exact formulas and model definitions were supplied where activated; the full catalog is not declared execution-ready.

R1.3

Change. Recorded actual provenance: six original author passes, six reviewer passes, five domain revision passes plus methods and integration work. Independent drafting is an intended recorded procedure, not a blanket retrospective claim.

Remaining. Detailed pass identifiers, timestamps, input visibility, and draft diffs would still be required for a methods evaluation. Same-workflow review is not external validation.

R1.4

Change. Integrated and schema-checked target statements, restricted questions, baselines, informative outcomes and explicit bridges for all 100 cards.

Remaining. These compact statements do not replace primary-source hypotheses or a full literature audit before substantial work.

R1.5

Change. Defined a principal composition question and cheap parity control. The integrating agent immediately executed the bounded composition test and found exact discrepancy 3/8, falsifying fixed-error preservation. The revised next principal task designs a correlated 16-point support using explicit integer constraints, limited adversarial bank, resource caps, and honest provisional/unknown statuses.

Remaining. The correlated-support repair and parity control are proposed, not completed. Passing a finite bank does not certify all six-bit programs; the bridge to arbitrary-length derandomization remains open.

R1.6

Change. Integrated separate route/readiness and completed-result fields. Completed results are specifically recorded for cards 6,23,37,46,81; other routes remain unexecuted.

Remaining. Readiness is a proposed-protocol label, not a claim of executed computation or established novelty.

R2.1

Change. Revised #6 defines S(n)=n/2 for even n and (3n+1)/2 for odd n, specifies the exact twelve-step affine formula on each class modulo 4096, and states its descent inequality. The accelerated map T is defined separately. The previously completed fixed-T-time obstruction remains in completed_pilot and baseline; the unexecuted S-residue protocol has its own explicit status and execution_status=calibration_ready.

Remaining. The residue inventory has not been executed. Even correct finite certificates would need a proof excluding every exceptional infinite trajectory before implying Collatz. No novelty or universal descent claim is made.

R2.2

Change. Revised #21 defines the height-bounded set of 13-smooth integers m≤p-1 and gives the additional character-sum inequality that would actually contradict the all-small-primes-residue hypothesis. The full residue subgroup modulo 17 is a mandatory negative control. The card explicitly retains the distributional assumptions in Tao's conditional route and forbids a sumset search without a candidate lemma stating its extra hypotheses.

Remaining. No new cancellation inequality is known from this work. The fixed smoothness threshold 13 is only a finite implication audit; the full conjecture requires control in the p^ε regime for every ε>0.

R2.3

Change. Revised #23 requires the E2 trace-one local factor to be zero before any model fit and uses trace two as a comparison without assuming its constant is positive. A new exact calibration counted points at all good primes p≤1000: E2 has 166 included primes, no odd traces, five trace-zero events, and six trace-two events. E1 has 54 odd traces but no observed trace-one event, exposing the difference between an empirical zero and a proved obstruction. The standard-library script and full JSON are saved separately. Completed results appear in baseline/completed_pilot; the NEXT task is the unexecuted modular Galois-image model at m=2,3,6, with execution_status=protocol_design. The target source is now Chen-Jones-Serban, Introduction equation (1.1).

Remaining. Actual modular Galois-image certificates and the growing-resolution comparison remain unexecuted. The finite control supplies no Lang-Trotter asymptotic, and no positive constant is assumed merely because a trace is not excluded by parity.

R3.1

Change. Resolved the conflation of join transfer with the difficult identification step; supplied a concise elementary lemma and proof in hajos_transfer_lemma.txt.

Remaining. No general rule for transporting disjoint clique-minor branch sets through arbitrary nonadjacent identification has been designed or proved.

R3.2

Change. Resolved the implication that rank-four existence is open. The proposed finite experiment now measures plateau escape geometry inside a known solved baseline.

Remaining. The exchange landscape experiment has not run, and no theorem transferring its behavior to arbitrary rank is supplied.

R3.3

Change. Resolved the unsupported readiness claim by labeling the card protocol_design and making geometric and coloring verification prerequisites to gluing.

Remaining. The exact coordinate file, chosen module, associated number field, and independent coloring certificates still need selection and verification before a concrete gluing search can begin.

R3.4

Change. Resolved the unsupported assumption of three ready small constructions and the incorrect optimization statistic. A fully defined toy grid is explicitly separated from the cited asymptotic construction; the card remains protocol_design.

Remaining. Feasibility of an exact instance with at most 200 points has not been established. Toy pruning, actual construction generation, and all new statistics remain unexecuted.

R4.1

Change. Item 56 now targets structured certificate extraction from Theorem A and Section 3's five-dimensional comparison, including mapping-class, framing, and Gluck-twist obligations. Its status is protocol_design. Geometric examples are explicitly calibration; any non-geometric extension is separately conditional.

Remaining. No certificate language, checker, or extracted proof has been completed. A non-geometric candidate needs precise admissibility and literature checks before selection; no new case or current-status audit is claimed.

R4.2

Change. Item 76 retains those examples only as regression controls and makes the real design task restoration of K-derivative terms in a source identity, followed by symbolic necessary-condition jet tests with Gauss-Codazzi and constant mean/scalar-curvature constraints.

Remaining. The identity, jet order, and candidate estimate still need to be selected and derived. Local jet feasibility is not global realizability, and passing such tests would not prove an integrated inequality or the conjecture.

R4.3

Change. Item 80 now cites that theorem, separates local calibration from finite-amplitude multistart discovery, identifies lattice and translation variables, and requires a compact nonsingular parameter domain plus certified neighbor cutoff before execution. Unrestricted three-point bounds remain a separate avenue.

Remaining. The precise parameter box, starting cells, cutoff proof, and optimization implementation are not yet designed. No local or nonlocal search has run; periodic search failure cannot establish the unrestricted packing bound.

R5.1

Change. Resolved the vacuous original test and completed its finite calibration. The revised research step is explicitly protocol design for a correlated sampler or a justified error-growth recurrence. These reported parent results were integrated, not independently rerun by this author.

Remaining. No growing-size sampler construction or general derandomization theorem follows. The composition obstruction concerns this fixed four-point support and independent concatenation only.

R5.2

Change. The invalid expectation of direct merges is removed. The replacement experiment is labeled protocol_design because a fixed ordered input decomposition, basis conventions, and search grammar still need code-level specification.

Remaining. No replacement search was executed. An empty screen or bounded failure cannot establish tensor-rank lower bounds, and a 48-term finite decomposition would not establish exponent two.

R5.3

Change. Resolved the finite-cutoff interpretation error analytically. Atom-exclusion is no longer an informative numerical outcome; any claimed separation must expose additional information or an error. The next cross-scale mechanism remains protocol_design.

Remaining. The finite-data obstruction neither refutes Furstenberg's conjecture nor excludes all cross-scale methods. No new cross-scale inequality, SDP execution, or limiting theorem is supplied.

R5.4

Change. Resolved the model ambiguity and advanced the finite comparison to calibration_ready. The source distinction between Maurer-style hidden handles and Shoup-style visible encodings is explicit. A copied handle-index bit is a required redundant-information control.

Remaining. The finite strategy enumeration is unexecuted. This artificial oracle cannot establish security for concrete encodings, unrestricted algorithm lower bounds, or existence of one-way functions.

R6.1

Change. Resolved in integration: run commands are separate preformatted lines; the combined command is absent. The pack includes a runnable verification orchestrator.

Remaining. Numerical reproduction and PDF rendering have separate dependencies, documented below.

R6.2

Change. Resolved in integration: the substantive descent statement now explicitly concerns odd n>1; the terminal-value fixture passed.

Remaining. The variable-time Collatz problem is untouched by this correction.

R6.3

Change. verification_controls.py and verification_results.json supplied, along with preserved baseline JSON. All 25 named checks passed.

Remaining. These controls and aggregate computations are not formal proof-assistant certificates or a per-case certificate archive; no such stronger claim is made.

R6.4

Change. Separate environment/hash manifest generated by the runnable verification script.

Remaining. Manifest covers numerical pilots and tests only; PDF-renderer dependencies are separate. Review remains an AI same-workflow follow-up; this reviewer previously authored the Collatz audit.

The revised 100-problem catalog

Every card now identifies its target, restricted question, baseline, informative outcome and missing bridge. The source links provide context or a stated result; they do not establish novelty for our suggested route. Definitions and exact hypotheses must be revisited when activating a project.

1-5 / Millennium problems

001 Riemann hypothesis

Simulated number theorist with analyst

Target. For the meromorphic continuation of the Riemann zeta function, every zero with $0 < \mathrm{Re}(s) < 1$ has $\mathrm{Re}(s)=1/2$. Define $\xi(s)=s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$, an entire function with precisely these zeros.

Restricted question. Can a polynomial approximation to ξ produce a rigorously certified zero count in the rectangle $0\le\mathrm{Re}(s)\le1$, $10\le\mathrm{Im}(s)\le20$, with a completely explicit boundary error budget?

Approach. Construct a Taylor polynomial centered at $1/2+15i$ and certify a uniform remainder bound on the rectangle boundary. Compare the polynomial winding number with ξ using a strict Rouché inequality. Record which derivative estimates, precision requirements, and subdivision counts deteriorate when the upper boundary is moved.

Challenge. A contour count neither places every counted zero on the critical line nor controls all heights. Symmetry permits off-line pairs; numerical polynomial roots alone do not certify the boundary inequality.

Next task. First specify polynomial degree, a rigorous remainder theorem, and interval winding-number computation. Attempt the single rectangle only after these ingredients yield checkable inequalities; retain failed bounds as output.

Baseline. Compare with a published finite-height zero-count certificate, recording its conventions and theorem location. This benchmark is established territory; the exact implementation reference remains to be selected.

Informative outcome. A certified count with explicit positive boundary margin calibrates the method; inability to separate remainder from boundary minimum identifies the first concrete obstruction. Neither outcome constitutes a new zeta theorem.

Bridge still needed. Still missing are control at unbounded height and a mechanism proving critical-line location. Improving a finite count is not such a mechanism; any claimed implication must be written separately.

Established / proposed. Completed-zeta methods, Rouché comparisons, and finite zero counts are established. This is a proposed error-budget audit; novelty and the closest prior implementation remain unchecked.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.claymath.org/millennium/riemann-hypothesis/

002 P versus NP

Simulated computational researcher with logician

Target. Does every language decidable by a nondeterministic polynomial-time Turing machine also have a deterministic polynomial-time decision algorithm? Equivalently, is P=NP? The alternative sought is a proved separation, not an experimentally slow solver.

Restricted question. For three eight-variable Boolean functions, how does the number of distinct residual functions change across all variable orders in an ordered binary decision diagram?

Approach. Use parity of eight bits, equality between the first and last four-bit blocks, and the conjunction of four adjacent two-variable disjunctions. At every cut of each variable order, substitute every prefix assignment and represent the residual function by its exact truth table. Compare maximum widths and total distinct residual states.

Challenge. A large width is a lower bound only for this ordered representation. Other algorithms or representations may bypass it; a finite test cannot imply asymptotic hardness or a separation of P and NP.

Next task. Enumerate all 40,320 permutations and every prefix assignment for each function, caching residual truth tables. Verify two selected orders by direct substitution before producing per-function minimum and maximum width tables.

Baseline. Use direct 256-entry truth tables for each function. Parity provides a simple control with at most two residual truth functions at every cut; the equality function tests sensitivity to paired versus separated ordering.

Informative outcome. A reproducible width table identifies whether an asserted order-independent obstruction survives these exact examples. An unexpectedly small order falsifies that candidate obstruction; a software discrepancy halts interpretation until repaired.

Bridge still needed. One would still need an infinite family, a proved growth bound, and a justified route from the restricted model to unrestricted polynomial-time computation. No such route is currently supplied.

Established / proposed. Decision diagrams and residual-function counting are established. The experiment is a calibration of representation dependence; no new lower bound or novel algorithm is claimed.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.claymath.org/millennium/p-vs-np/

003 Birch and Swinnerton-Dyer conjecture

Simulated algebraist and arithmetic geometer

Target. For every elliptic curve E/Q, BSD predicts ord_(s=1)L(E,s)=rank E(Q)=r, finiteness of Sha(E/Q), and lim_(s→1)L(E,s)/(s-1)^r=Ω_E Reg_E |Sha| ∏_p c_p / |E(Q)_tors|². Use the global minimal differential for Ω_E=∫E(R)|ω| and regulator convention 1 at rank zero.

Restricted question. For E:$y^2=x^3-x$, can we reconstruct a documented rank-zero benchmark and separately audit every factor in the leading-coefficient formula, distinguishing certified inputs from numerical estimates?

Approach. Create two dependency ledgers: one for rank equality and another for the full leading coefficient. Record rational points, descent bounds, torsion, local component numbers, real period, analytic evaluation, and the status of Sha. Never estimate Sha by rearranging the conjectural formula and then count that estimate as verification.

Challenge. A point search gives no upper rank bound, a tiny L-value does not certify its vanishing order, and assuming BSD inside a database routine would make the comparison circular.

Next task. Find the primary benchmark calculation and document its exact hypotheses. Reproduce the torsion and local-factor inputs, then classify each remaining quantity as proved, conditional, or numerical before making any equality claim.

Baseline. This classical curve is a solved calibration target, not a proposed new BSD case. Compare independently justified rank-zero and torsion statements with the selected source, documenting period and component conventions.

Informative outcome. A complete convention-consistent ledger is useful preparation. A missing Sha justification or numerical interval containing zero is an explicit unresolved dependency, not a discrepancy against the conjecture.

Bridge still needed. Neither a solved rank-zero case nor arithmetic bounds for a few curves address all ranks or the full leading coefficient. General rank control and the unproved dependencies must remain separate targets.

Established / proposed. All listed BSD quantities and benchmark methods are established; this is an unexecuted dependency audit without a novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.claymath.org/millennium/birch-and-swinnerton-dyer-conjecture/

004 Hodge conjecture

Simulated algebraic geometer with topologist

Target. For every smooth projective complex variety X and every p≥0, each rational cohomology class in H^(2p)(X,Q) whose complexification has Hodge type (p,p) is a rational linear combination of classes of codimension-p algebraic cycles.

Restricted question. On X=E×E for E:$y^2=x^3-x$ over C, can we explicitly match a basis of rational degree-two Hodge classes with divisor classes, including the extra class supplied by complex multiplication?

Approach. Use the horizontal and vertical elliptic curves, the diagonal, and the graph of the automorphism i:(x,y)↦(-x,iy). Compute their intersection matrix and cohomological actions in a chosen product basis. Compare their span with the rational (1,1) subspace, retaining the role of complex multiplication explicitly.

Challenge. The divisor case is already established. A complete calculation on a product does not extend to higher codimension, and omitting the extra endomorphism would produce an incomplete basis rather than evidence against Hodge.

Next task. Fix oriented integral cycle bases on both factors; calculate the four divisors' pairwise intersections and exact cohomology coordinates, then verify independence and equality of their span with the identified rational Hodge subspace.

Baseline. Use the established divisor case and the product cohomology decomposition. Check the graph equation directly and reconcile the intersection calculation with the degree of the difference between the identity and multiplication by i.

Informative outcome. A matching basis certifies this training calculation. A dimension mismatch forces correction of the Hodge subspace, basis conventions, or missing cycles before any speculative extension is considered.

Bridge still needed. A separate construction is needed in higher codimension on varieties without product decomposition. Nothing here supplies that construction; its hypotheses and the first unavailable correspondence must be stated before advancing.

Established / proposed. Divisor algebraicity, product cohomology, and complex multiplication are established. The task is an explicit calibration, with no new Hodge case or novel construction asserted.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.claymath.org/millennium/hodge-conjecture/

005 Yang-Mills existence and mass gap

Simulated analyst with probabilist and mathematical-physics perspective

Target. For every compact simple gauge group G, construct a nontrivial quantum Yang-Mills theory on four-dimensional Euclidean space satisfying the rigor specified in the official problem, and prove a strictly positive mass gap Δ above its vacuum in the reconstructed physical theory.

Restricted question. In one explicitly selected lattice approximation, which constants in a finite-system spectral-gap estimate depend on spacing, coupling, and volume, and which would have to survive removal of the cutoff?

Approach. Choose a published finite-lattice inequality and rewrite its proof as a parameter ledger, including normalization of physical energy. Track changes under increased box size and decreased spacing separately. Identify whether the estimate concerns a transfer operator, a physical Hamiltonian, or merely an auxiliary sampling process before interpreting its gap.

Challenge. A sampling-chain gap is not a physical mass gap. Finite-volume positivity can disappear after scaling or limiting operations, and a surviving inequality alone does not construct a continuum field theory satisfying the required axioms.

Next task. Select and cite one exact theorem with lattice action, boundary conditions, coupling regime, and gap definition. Reproduce its finite-system calculation before attempting any numerical comparison or extrapolation; no unselected model counts as execution-ready.

Baseline. The chosen theorem itself supplies the baseline; its primary source and hypotheses must be recorded before calculation. An elementary finite operator may calibrate linear algebra but cannot substitute for the specified gauge-theory model.

Informative outcome. A checked dependence table can reveal a missing uniformity estimate, an incompatible scaling regime, or a gap defined for the wrong operator. Each is an informative reason to reformulate the route.

Bridge still needed. Still required are continuum construction, compatibility of limiting operations, verification of field-theory axioms, and positivity of the physical gap. These are substantive mathematical problems, not consequences of finite-lattice measurements.

Established / proposed. Lattice approximations and gap estimates are established tools. This proposal audits their exact scope; no physical construction, universal estimate, or originality is claimed.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.claymath.org/millennium/yang-mills-the-maths-gap/

6-32 / Number theory

006 Collatz conjecture

Simulated team: dynamical-systems researcher

Target. Every positive integer reaches 1 under C(n)=n/2 for even n and C(n)=3n+1 for odd n.

Restricted question. Which residue classes admit a certified decrease after exactly twelve one-division shortcut steps, and what inequalities remain unresolved?

Approach. Use S(n)=n/2 for even n and S(n)=(3n+1)/2 for odd n. For each r modulo 4096, derive S^12(n)=(3^a n+b)/4096 on n≡r. Certify descent only where (4096-3^a)n>b. Preserve exceptional small inputs and unresolved expanding classes explicitly.

Challenge. These are shortcut steps, not accelerated odd steps T(n)=(3n+1)/2^v2(3n+1). A fixed residue modulus does not determine twelve T-branches. Moreover, a finite graph can retain an infinite exceptional path.

Next task. Enumerate r=0,...,4095, compute each parity word and exact integers a,b, and independently check the affine identity at three positive representatives. Spend at most two CPU hours; output inequalities and unresolved classes, without labeling graph coverage as universal descent.

Baseline. Direct integer iteration validates formulas. Include the completed T-map obstruction: starting at 2^m-1 gives m-1 initial increases, so universal fixed accelerated descent time is impossible.

Informative outcome. An exact error in a branch formula rejects the implementation. A surviving expanding class redirects work toward adaptive certificates; an attractive resolved percentage alone does not justify another larger scan.

Bridge still needed. Unknown: prove every positive trajectory eventually enters a certified descending class, including all exceptional paths, then use induction. The existing fixed-time obstruction does not supply that theorem.

Established / proposed. Affine residue analysis is established. The proposed certificate inventory is a reproducibility exercise; no novelty is asserted.

Proposed route. calibration ready; readiness does not imply execution.

Completed result. Retained result: for every k>=1 there is an odd n>1 whose first k accelerated odd steps increase. The formula for n=2^m-1 is proved below; the finite audit is reproduced. The shortcut residue-graph proposal is separate and unexecuted.

https://arxiv.org/abs/1909.03562

007 Goldbach conjecture

Simulated team: additive-number-theory researcher

Target. Every even integer N≥4 is a sum of two primes, allowing the two primes to be equal.

Restricted question. Do unusually small exact representation counts retain a common residual feature after accounting for elementary local divisibility factors?

Approach. Define the ordered count R(N)=Σ_(2≤p≤N-2)1_prime(p)1_prime(N-p). Separate it from B_y(N)=S_y(N)Σ_(2≤m≤N-2)1/[log(m)log(N-m)], where S_y(N)=∏_(ℓ≤y)(1-ν_ℓ(N)/ℓ)/(1-1/ℓ)^2 and ν counts forbidden residues 0,N modulo ℓ. Inspect residuals before designing smoothing.

Challenge. The truncated product has no justified tail bound here, especially at small N. Averaging neighboring integers can improve a residual plot while removing the pointwise positivity that Goldbach needs.

Next task. Use a sieve through 200000, exact ordered counts for even N, and y=7,31,97. Rank R(N)/B_y(N) on N≥1000; compare residue classes modulo 210. The smoothing kernel and a rigorous residual inequality still require design, so do not activate an optimization run.

Baseline. Verify R(4)=1 and account for ordered off-diagonal pairs twice. Compare local correction against the same logarithmic sum with S_y=1, using fixed cutoffs rather than fitted constants. Export the normalization choices alongside every count.

Informative outcome. A lowest-count stratum stable across cutoffs identifies a concrete analytical case to explain. If the ranking is dominated by truncation choices or small-N effects, reject that feature before constructing a kernel.

Bridge still needed. Unknown: a uniform unsmoothed error bound strong enough to prove R(N)>0 for every sufficiently large even N, followed by a certified finite check of the remainder.

Established / proposed. Local factors and circle-method heuristics are established context. This residual comparison is proposed diagnostic work, not a new Goldbach estimate.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://t5k.org/notes/conjectures/

008 Twin prime conjecture

Simulated team: sieve-methods researcher

Target. There are infinitely many primes p for which p+2 is also prime.

Restricted question. Do three fixed, primality-blind divisor weights separate prime pairs from composite-containing pairs after normalization and an unchanged holdout test?

Approach. Define W_j(n;R)=[Σ_(d≤R,d|n(n+2)) μ(d)(1-d/R)^j]^2 for j=1,2,3, giving all three promised formulas. Here μ is the Möbius function. These elementary rational tapers are chosen for exact auditing, not asserted to be optimal Selberg weights. Primality enters only subsequent outcome labels.

Challenge. A positive square weights composites too. Better empirical separation does not create a lower-bound sieve for twin primes, and tuning a taper after seeing primality labels would undermine this diagnostic.

Next task. Fix R=30,100. Compute all six weights on 2≤n≤20000, then unchanged on 20001≤n≤40000. Use integer numerators [Σ μ(d)(R-d)^j]^2 with denominator R^(2j). Record total mass, prime-pair mass, and masses by Ω(n)+Ω(n+2), counting factors with multiplicity. Cap execution at two CPU hours.

Baseline. Compare normalized prime-pair mass with W=1 and the simple indicator gcd(n(n+2),30)=1. A separate sieve supplies exact primality labels and Ω values; neither modifies the weights.

Informative outcome. A gain that disappears on the holdout interval or changes sign with R parks this family. Stable improvement earns an algebraic decomposition of composite mass, not a claim that the parity barrier has been passed.

Bridge still needed. Unknown: estimates for the weight sums and composite contributions strong enough to force an unbounded count of genuine prime pairs. Finite enrichment alone supplies none of these estimates.

Established / proposed. Goldston-Pintz-Yıldırım, Primes in Tuples I, equation (2.13), provides a nearby established logarithmic divisor-sum family. Our rational tapers differ deliberately; their audit carries no novelty claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/math/0508185

009 Legendre’s conjecture

Simulated team: analytic-number-theory researcher

Target. For every integer $n \geq 1$, at least one prime p satisfies $n^2 < p < (n+1)^2$.

Restricted question. Which of three compactly supported polynomial kernels offers the best measured interior prime mass relative to its explicitly computed derivative norms?

Approach. Put $t=(m-n^2)/(2n+1)$ and use $w_j(t)=c_j[t(1-t)]^j$ on $0<t<1$, zero otherwise, for $j=2,3,4$; choose c_j so the integral is one. Compare weighted von Mangoldt sums with their interior prime-only contributions. Their prime-power difference must remain visible in any eventual positivity argument.

Challenge. A positive von Mangoldt sum can count a higher prime power rather than a prime. Kernel regularity helps transform decay but supplies no uniform bound for the contribution of zeta zeros.

Next task. For $1 \leq n \leq 2000$, sieve through 2001^2 and tabulate the three weighted sums. Derive normalization and derivative integrals symbolically; use certified logarithm intervals for comparisons. Before numerical optimization, write the exact explicit-formula remainder and prime-power subtraction required for each kernel.

Baseline. Use the unweighted prime count in each interval as ground truth. The same kernel applied to all integers measures deterministic discretization error separately from prime-distribution error.

Informative outcome. Reject a kernel comparison if its apparent advantage comes from prime powers or boundary normalization. A useful output is a precise missing zero-sum inequality with a quantified required constant, not a nicer graph.

Bridge still needed. Unknown: prove the weighted prime-only main term exceeds all errors uniformly at interval length $2n+1$. Finite kernel comparisons cannot transfer that guarantee to every n.

Established / proposed. Weighted explicit formulas and prime-power corrections are standard. This limited kernel audit is proposed work with no claimed new short-interval theorem.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://t5k.org/notes/conjectures/

010 Landau’s n^2+1 problem

Simulated team: arithmetic-geometry researcher

Target. Infinitely many positive integers n have n^2+1 prime.

Restricted question. After explicit congruence corrections, does a widening-strip comparison isolate any estimate genuinely uniform down to the fixed Gaussian-integer line n+i?

Approach. Write $n^2+b^2=N(n+bi)$, but keep the original integer prime condition rather than counting all Gaussian primes indiscriminately. Compare b=1,2,3 and cumulative strips $1\le b\le B$ for B=1,2,4,8. Separate the gain from having more lines from any improvement in control of the single target line.

Challenge. A strip average can be accurate while the line b=1 remains uncontrolled. Common factors of n and b force composite norms, and ignoring these produces artificial differences between the lines.

Next task. For $n\le 20000$, record exact prime norms and gcd(n,b) for each line. For each b, compute roots of x^2+b^2 modulo primes at most 31 and use those root counts to state a local baseline. Specify a uniform strip-error inequality before describing the resulting counts as an analytic test.

Baseline. Use exact ordinary-integer primality as ground truth. Include pairs with gcd(n,b)>1 as forced-composite controls, and separate exceptional small values when a norm equals the sieving prime itself.

Informative outcome. If normalized strip agreement improves only because B grows, report an averaging effect and park the strip-to-line inference. Retain a mechanism only if it identifies a quantitatively meaningful error term at B=1.

Bridge still needed. Unknown: obtain an unbounded prime count on b=1 with error smaller than its expected main term. Sector or strip theorems lacking this uniformity do not imply the target.

Established / proposed. Gaussian norms and local polynomial obstructions are established. The revised proposal audits a change of viewpoint; it does not claim that the norm representation resolves prime detection.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1807.08899

011 Infinitely many Mersenne primes

Simulated team: computational-number-theory researcher

Target. Infinitely many prime exponents p make the Mersenne number 2^p-1 prime.

Restricted question. How much of compositeness among prime exponents p≤1000 is explained by fully enumerated trial factors q≤1000000?

Approach. For each candidate exponent, compare exact Lucas-Lehmer primality with a bounded factor sieve. If q divides 2^p-1 then ord_q(2)=p for odd p, so a fixed prime divisor excludes that single prime exponent, not a large family of unrelated prime exponents. Study families of possible q without inventing correlations from this identity.

Challenge. No bounded factor sieve can certify primality merely by survival. Prime divisors above the cutoff can account for every remaining composite candidate; a density argument must control that entire missing tail.

Next task. Enumerate all prime p≤1000 and prime q≤1000000. Test divisibility with modular exponentiation, record exact orders only for actual factors, and compare with Lucas-Lehmer residues. Handle p=2 separately. Freeze both cutoffs before computing and distinguish certified composite survivors from certified Mersenne primes.

Baseline. The Lucas-Lehmer test is a known complete primality criterion for prime exponents. Its outcome calibrates the incomplete trial sieve, while small examples such as p=11 verify that a composite Mersenne number can have a short certificate.

Informative outcome. Report the fraction of known composites missed by each nested factor cutoff and their least discovered factors. If a proposed covering statistic simply counts one excluded exponent per actual divisor, discard any claimed cross-exponent independence conclusion.

Bridge still needed. Unknown: an argument leaving infinitely many exponents free of every possible divisor, including factors far beyond the experimental cutoff. Observed survival density does not establish such a statement.

Established / proposed. Order restrictions and Lucas-Lehmer are established tools. The finite audit is reproducibility work; the infinite covering mechanism still requires design.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://t5k.org/mersenne/

012 Odd perfect number problem

Simulated team: Diophantine-equations researcher

Target. There is no odd positive integer N with σ(N)=2N, where σ is the sum of positive divisors.

Restricted question. Can a depth-four factor-chain database produce reusable, independently checkable pruning certificates without suppressing unbounded exponents or unlisted primes?

Approach. Use Euler form N=q^a m² with q prime, gcd(q,m)=1 and q≡a≡1 mod 4. Each explored nonspecial prime power p^e has even e, and odd prime factors of σ(p^e) must divide N. Store forced divisibility and exact partial abundancy separately from heuristic choices of subsequent exponents.

Challenge. The special prime may be absent from a short branch, and its exponent cannot be treated as even. Cyclic forcing does not itself contradict perfection; a branch cut off by depth or factorization limits remains unresolved.

Next task. Explore starts p=3,5,7,11 with nonspecial exponents e=2,4,6, allowing a separate special-prime state wherever compatible. Factor each encountered σ(p^e) exactly through depth four. Prune when a certified divisor d of N satisfies σ(d)/d>2; explicitly retain every other unexplored exponent and continuation.

Baseline. Cross-check each σ value from its geometric-series formula. Verify monotonicity of the partial abundancy bound under extending prime exponents, and run the certificate checker independently of branch-generation order.

Informative outcome. A useful result is one sound reusable pruning lemma with its complete hypotheses. If repeated states omit special-prime information or exponent ranges, reject their proposed compression even when many finite branches vanish.

Bridge still needed. Unknown: finite certificates covering every admissible Euler factorization or a well-founded argument exhausting the infinite search tree. A depth-four census cannot exclude all odd perfect numbers.

Established / proposed. Euler form, factor chains, and abundancy bounds are established. Compression is a proposed organizational experiment without novelty or completeness claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2607.19746

013 Beal conjecture

Simulated team: modular-arithmetic researcher

Target. If positive integers A,B,C and integers x,y,z>2 satisfy A^x+B^y=C^z, then gcd(A,B,C)>1.

Restricted question. Which local residue patterns survive the exponent box 3≤x,y,z≤7, and can every exclusion be represented by a short exact certificate?

Approach. Treat primitive candidate solutions as a constraint system, preserving valuations as well as unit residues. For each prime modulus, a primitive solution cannot have all three bases divisible by that prime. Within a solution, divisibility of two bases forces divisibility of the third; enforce this necessary implication explicitly.

Challenge. Survival at finitely many moduli establishes only local compatibility. Different residue tuples can be combined by the Chinese remainder theorem without yielding an integer solution, and local failure in a bounded exponent box is not a universal exponent argument.

Next task. For all 125 exponent triples, enumerate residues modulo 16,27,25,7,11,13. Keep zero residues where primitiveness permits them; export allowed tuples and the precise failed congruence for each exclusion. Design a compact compatibility representation before attempting any combined-modulus enumeration.

Baseline. Check the nonprimitive equality $3^3+6^3=3^5$ as an equation control that the primitive filter must remove. Preserve unit-residue solutions of isolated congruences even when no global equality is known.

Informative outcome. An independently checked exclusion for an exponent class is useful. If all classes remain locally admissible, document the surviving patterns and stop adding small primes without a new global mechanism.

Bridge still needed. Unknown: control all exponent triples and rule out globally compatible primitive integer solutions. Modular certificates could support a restricted theorem, but the remaining step needs additional arithmetic information.

Established / proposed. Valuation arguments and modular sieves are established. This certificate map is proposed bookkeeping, not a claimed new proof method or a replacement for global Diophantine techniques.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.ams.org/profession/prizes-awards/ams-supported/beal-prize

014 Erdős-Moser conjecture

Simulated team: algebraic-number-theory researcher

Target. The only positive integers m≥2 and k≥1 satisfying 1^k+⋯+(m-1)^k=m^k are m=3,k=1, giving 1+2=3.

Restricted question. For even exponents 2≤k≤60, which congruence restrictions modulo small prime squares survive exact simultaneous compatibility checks?

Approach. Compute F_k(m)=Σ_(j=1)^(m-1)j^k-m^k modulo p^2. First prove that p^4 is a valid period in m: shifting by p^4 adds p^2 complete blocks of length p^2, each with the same residue sum. Only after this elementary certification should residue tables be used as exhaustive local information.

Challenge. Combining independent moduli often preserves many classes by the Chinese remainder theorem. A large lower bound on a possible solution or failure within these thirty exponents does not handle arbitrary k.

Next task. For p=2,3,5,7,11,13, update power sums incrementally over $0≤m<p^4$, using exact modular arithmetic for each even k. Keep the congruence classes as symbolic CRT products to avoid a huge expanded table. Separately check the genuine solution (m,k)=(3,1).

Baseline. Compare incremental updates against direct sums for m≤100. Verify the period identity symbolically and test boundary residues m=0,1,p^2-1 rather than silently assuming m is a unit.

Informative outcome. Seek a short inconsistency certificate within one exponent class or a clear demonstration that independent congruences do not interact. If all combinations survive, stop widening the prime list until a shared arithmetic condition is identified.

Bridge still needed. Unknown: a uniform restriction on unbounded exponents and candidate m that forces a contradiction. This finite CRT study can expose a bottleneck but cannot close the global equation.

Established / proposed. Power-sum congruences and Moser-style restrictions are established. The period check and compatibility inventory are calibration work without a novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1011.2940

015 Bunyakovsky conjecture

Simulated team: polynomial-primes researcher

Target. Every nonconstant irreducible $f \in \mathbb{Z}[x]$ with positive leading coefficient and no prime dividing every value $f(n)$ takes prime values for infinitely many positive integers n.

Restricted question. Can any fully specified small-divisor statistic distinguish three admissible irreducible examples from reducible controls without using primality information?

Approach. Use $f = x^2+1, x^2+x+41, x^3+2$ and controls $g = (x+1)(x+2), (x+1)(x^2+1)$. Separate irreducibility certificates, fixed-prime-divisor tests, and observed factorization parity. These controls are deliberately reducible but are not claimed to match all local factors; any matching procedure must be specified before making such a comparison.

Challenge. A statistic can distinguish coefficient patterns or obvious local obstructions while saying nothing about prime detection. Matching finitely many local factors does not distinguish primes from integers with several large factors.

Next task. For $1 \le n \le 10000$, factor the five polynomial values exactly and record Ω, sieve survival for primes at most 31, and primality as an outcome label only. Compute all corresponding root counts modulo those primes. Design any stratification or matching rule on these local counts before testing separation.

Baseline. For each polynomial, test a complete residue system modulo every prime up to its degree to exclude a fixed prime divisor; for larger primes the root-count bound applies once content and reduction are checked. Independently verify irreducibility over Q.

Informative outcome. A difference explained entirely by local root counts rejects the proposed global statistic. A surviving distinction earns an explicit formula and a theorem-sized question; finite classification accuracy alone is not progress on Bunyakovsky.

Bridge still needed. Unknown: a uniform prime-detection estimate for every fixed admissible irreducible polynomial. Neither factorization tables nor matched local data supply a positive lower bound for its prime values.

Established / proposed. Polynomial sieves and their parity limitations are established. This controls-first revision is methodological and claims no new criterion for polynomial prime values.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1807.08899

016 Bateman-Horn conjecture

Simulated team: probabilistic-number-theory researcher

Target. For distinct irreducible $f_1,\dots,f_k \in \mathbb{Z}[x]$ with positive leading coefficients and no fixed prime divisor of their product, the joint prime count is asymptotic to $C_f \int_T^X dt/\prod_i \log f_i(t)$. Here T makes all $f_i(t)>1$, and $C_f=\prod_\ell (1-\nu_\ell/\ell)/(1-1/\ell)^k$, with ν_ℓ the number of roots of $\prod_i f_i$ modulo ℓ.

Restricted question. For $f_1(t)=t$ and $f_2(t)=t^2+a$, how sensitive is the predicted finite count to truncating the Euler product before comparing different a?

Approach. Fix $a=4+6j$ for $j=0,\dots,9$. Define $\nu_\ell(a)=\#\{t \bmod \ell : t(t^2+a)=0\}$ and $C_y(a)=\prod_{(\ell\le y)}(1-\nu_\ell(a)/\ell)/(1-1/\ell)^2$. Compare counts with $C_y(a)\int_2^X dt/[\log(t)\log(t^2+a)]$, preserving coefficient dependence and the same integral convention for every pair.

Challenge. The finite products need not approximate their limits uniformly at these cutoffs. A visually successful coefficient average can hide a poorly predicted fixed polynomial; fitting a separate constant to each curve would make the comparison circular.

Next task. Use $X=1000,3000,10000$ and $y=31,101,1009$. Sieve exact prime values, enumerate root counts directly, and evaluate integrals with certified error intervals. Before execution, select a computable tail bound or explicitly mark the constant error uncontrolled; no extrapolated asymptotic fit is authorized.

Baseline. Prove admissibility: modulo 2 and 3 the chosen a leave allowed residues, and primes at least 5 cannot be covered by at most three roots. Compare each truncated product with the uncorrected integral.

Informative outcome. A rank reversal between coefficients as y increases identifies constant truncation as the immediate bottleneck. Stable discrepancies justify investigating one fixed pair, not reporting an averaged confirmation of the conjecture.

Bridge still needed. Unknown: establish the Euler-product normalization and prime-count asymptotic for each fixed admissible family, with an error smaller than the main term. A coefficient average is a different theorem.

Established / proposed. Bateman-Horn local factors are established conjectural machinery. This sensitivity audit is a proposed reproduction exercise without novelty claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1807.08899

017 Artin’s primitive root conjecture

Simulated team: Galois-theory researcher

Target. For every integer g with |g|>1 that is not a perfect square, infinitely many primes p∤g satisfy ord_p(g)=p-1.

Restricted question. For g=2,3,5, what proportion of nonprimitive primes has every prime divisor of the order index larger than 13?

Approach. Set I_g(p)=(p-1)/ord_p(g). A prime ℓ divides this index exactly when g^((p-1)/ℓ)≡1 mod p with ℓ|p-1. Separate failures witnessed by ℓ≤13 from those surviving every small-index test; report dependence using exact intersections rather than assuming the obstruction events are independent.

Challenge. Computing all finite order indices does not bound the large-index tail as p grows. Inclusion-exclusion over small primes can look accurate while rare failures dominate the missing asymptotic argument.

Next task. Sieve p and factor p-1 for all primes p≤1000000, excluding p|g. Compute orders by repeated prime-factor removal from p-1 and verify each result by modular powers. Tabulate obstruction intersections for ℓ=2,3,5,7,11,13, splitting p into fixed logarithmic-size bands.

Baseline. Use bases g=1 and g=4 as intentionally excluded controls: neither should produce primitive roots at odd primes. Compare actual intersections with the product of measured marginal frequencies, labeling the latter an empirical independence model only.

Informative outcome. The deliverable is the exact family of failures not explained by small indices and a proposed tail inequality with quantified dependence on g and the cutoff. If no such inequality emerges, park further order enumeration.

Bridge still needed. Unknown: control the large-index obstruction tail uniformly enough to leave a positive-density or merely infinite primitive-root set. Conditional results cannot be silently treated as unconditional.

Established / proposed. Order-index tests and inclusion-exclusion are established. This revision specifies controls and the missing tail statement; it asserts neither a new density formula nor an Artin proof.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/math/0412262

018 Elliott-Halberstam conjecture

Simulated team: harmonic-analysis researcher

Target. For every $0<\theta<1$ and $A>0$, $\Sigma_{q\le x^\theta} \max_{(a,q)=1}|\psi(x;q,a)-x/\varphi(q)| \ll_{A,\theta} x/(\log x)^A$, where ψ sums the von Mangoldt function in a residue class.

Restricted question. Which largest-prime-factor bands dominate this exact finite progression-error sum when $\theta=0.7$?

Approach. Retain the maximum over reduced residue classes for each modulus; replacing it by an average would change the target. Partition $q\ge2$ into $P^+(q)\le13$, $13<P^+(q)\le\sqrt{q}$, and $P^+(q)>\max(13,\sqrt{q})$. Treat $q=1$ separately and keep signed errors available behind each absolute maximum.

Challenge. At $x=10000$ or 100000, boundary terms and prime powers can dominate an asymptotic pattern. Improvements for smooth moduli do not automatically control the complementary band, even if it contains fewer moduli.

Next task. For both x values, use q satisfying $q^{10}\le x^7$. Enumerate all prime powers through x, accumulating $\Lambda(n)=\log p$ with certified intervals, then compute every reduced-class error. Specify a dispersion identity for the intended extension before interpreting bands as evidence for an analytic gain.

Baseline. Verify that summing ψ over reduced classes equals the total prime-power weight coprime to q. Compare the three-band total with a direct unsplit computation to prevent accidental omission of difficult moduli.

Informative outcome. A hard band retaining a comparable error contribution rejects any proposal to discard it by cardinality alone. A useful result identifies the exact term a factorization argument must bound rather than simply the smallest observed band.

Bridge still needed. Unknown: uniform dispersion bounds for all moduli up to every exponent $\theta<1$ and arbitrary logarithmic savings. Two finite x values cannot establish those quantifiers.

Established / proposed. Progression errors and smooth-modulus decompositions are established. This band audit is proposed diagnostic work and does not upgrade restricted distribution results to Elliott-Halberstam.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1407.4897

019 Chowla’s conjecture

Simulated team: multiplicative-functions researcher

Target. For every k≥1 and distinct fixed nonnegative shifts $h_1,\dots,h_k$, $\Sigma_{n\le X}\prod_j \lambda(n+h_j)=o(X)$, where $\lambda(n)=(-1)^{\Omega(n)}$.

Restricted question. For two-point correlations, which apparent sign biases remain after the exact contribution of primes at most 7 has been separated?

Approach. Write n=n_small n_rough, where n_small contains all prime powers with prime at most 7, and λ(n)=λ(n_small)λ(n_rough). Compare the original λ(n)λ(n+h) with the rough-part product for h=1,...,20. Conditioning is an observational decomposition, not an independence assumption or a substitute for the full unweighted conjecture.

Challenge. Removing small primes can increase conditional bias and does not remove the difficult parity information. Results for logarithmically weighted correlations or special orders must not be described as the unrestricted unweighted statement.

Next task. Sieve Ω and least prime factors through 1000020. Use windows [1,10000], [1,100000], [1,1000000], stratified by n mod 30, and record exact signed sums plus sample sizes. Choose no data-dependent shifts or strata; any proposed cancellation inequality must be written separately before further computation.

Baseline. Reconstruct λ(n) exactly from its two factors for every input. Include shift h=0 as a known noncancelling control, whose original product is identically one; do not place that control inside the conjecture.

Informative outcome. A conditional bias larger than the unconditioned bias falsifies a proposed monotonic improvement from removing small primes. Stable cancellation merely motivates a precisely stated conditional estimate; it does not validate independence.

Bridge still needed. Unknown: an unweighted correlation estimate uniform enough in all fixed distinct shifts and every order k. Two-point finite diagnostics address only one restricted corner of that requirement.

Established / proposed. Liouville correlations and small-prime factorization are established. The explicit control and conditioning audit are proposed methodological work without a novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1509.05422

020 Lindelöf hypothesis

Simulated team: complex-analysis researcher

Target. For every ε>0, |ζ(1/2+it)|=O_ε((1+|t|)^ε) as |t| tends to infinity.

Restricted question. What quantitative peak-width bound would make a chosen mean-value estimate imply a stronger pointwise bound, and does the proposed derivative argument actually prove it?

Approach. Start symbolically: if F(t)=|ζ(1/2+it)| has height H and an upper Lipschitz bound D on a surrounding interval, then F≥H/2 for a width of at least H/(2D), provided that interval is available. Thus its 2k-th moment contributes at least a constant times H^(2k+1)/D. Keep the dependence of D on H and t explicit.

Challenge. The derivative bound may already be as difficult as the desired pointwise estimate. A sampled derivative maximum is not a rigorous upper bound between samples, and circular substitution can make the moment argument look stronger than it is.

Next task. Before sampling, choose one published moment inequality and one rigorously applicable derivative bound, stating constants, interval lengths, and hypotheses. Derive the resulting exponent algebraically. Only if it improves a known bound should heights 100≤t≤1000 be used to test numerical implementation.

Baseline. Use a triangular Lipschitz peak to verify the elementary width-to-moment inequality and endpoint effects. Compare the derived exponent with the same mean-value estimate without the proposed improvement.

Informative outcome. If substituting the derivative bound returns the starting exponent, record an exact circularity certificate and stop the numerical peak search. A useful advance is an independently justified derivative or width inequality with stronger dependence.

Bridge still needed. Unknown: sufficiently strong uniform peak control and moment estimates yielding every exponent ε>0. An isolated-height improvement would not establish Lindelöf.

Established / proposed. Moment and large-value arguments are established. This algebra-first revision is an implication audit, not a new zeta-function estimate.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2606.02407

021 Vinogradov’s least quadratic nonresidue conjecture

Simulated team: additive-combinatorics researcher

Target. For every $\varepsilon>0$ there is C_ε such that the least positive quadratic nonresidue $n(p)$ satisfies $n(p)\le C_\varepsilon p^{\varepsilon}$ for every odd prime p.

Restricted question. Can a proposed smooth-number argument supply character cancellation, rather than merely an expanding sumset, under the hypothetical condition that all small integers are residues?

Approach. Define $H=p-1$ and $S=\{m\le H:$ every prime divisor of m is at most $13\}$. If every prime at most 13 is a quadratic residue modulo p, then $\chi_p(m)=1$ for every $m\in S$. A contradiction requires a separate valid estimate such as $|\Sigma_{m\in S}\chi_p(m)|<|S|$. Sumset expansion alone supplies no such estimate.

Challenge. The missing character bound is the hard implication, not a bookkeeping detail. Indeed the full nonzero quadratic-residue set modulo 17 has sumset equal to the entire field while every original element remains a residue.

Next task. Within a two-hour audit, take the first 200 primes at least 10000. Record small-prime character patterns, construct the height-bounded S exactly, and verify the conditional identity for compatible primes. Independently reproduce the modulo-17 subgroup control. Do not launch a sumset search unless a candidate cancellation lemma states its additional hypotheses.

Baseline. The full residue subgroup is a mandatory negative control against any expansion-implies-contradiction rule. Tao’s cited paper offers a conditional distributional route; its Elliott-Halberstam or Type II assumptions must remain explicit.

Informative outcome. Reject any lemma contradicted by the subgroup control or satisfied only by assuming the desired cancellation. Retain a route only after its hypotheses exclude that control for a mathematically relevant reason.

Bridge still needed. Unknown: prove the necessary character estimate with a smoothness threshold comparable to p^{ε} for every ε. The fixed bound 13 is calibration, not the conjectured regime.

Established / proposed. Smooth numbers and character sums are established; the corrected implication audit asserts no new cancellation theorem.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1410.7073

022 Lehmer’s conjecture on Ramanujan’s tau function

Simulated team: modular-forms researcher

Target. For every positive integer n, τ(n)≠0, where q∏_(m≥1)(1-q^m)^24=Σ_(n≥1)τ(n)q^n.

Restricted question. What precise hypothetical-zero restrictions follow from the Hecke recurrence and the congruence τ(n)≡σ₁₁(n) mod 691 in a certified finite prime-power table?

Approach. For each prime p, use τ(p^(r+1))=τ(p)τ(p^r)-p^11τ(p^(r-1)), starting from τ(1)=1. Track recurrence consequences separately from residue exclusions. In particular, a hypothetical zero at p requires 1+p^11≡0 mod 691; do not confuse nonzero values of τ(p) modulo another prime with nondivisibility by p.

Challenge. Congruences leave infinitely many admissible primes, and recurrence propagation does not automatically prove all prime powers nonzero. The cited paper’s conditional statement about prime values of τ is not an unconditional nonvanishing theorem.

Next task. Design an exact q-series or modular-forms computation through n=50000, extract every prime power in that range, and check the recurrence against independently computed coefficients. Export each exclusion with its exact residue certificate and retain survivors. Identify the relevant theorem references before expanding the filter collection.

Baseline. Check τ(1)=1, τ(2)=-24, and τ(4)=-1472 both from the product and from the recurrence. Verify the 691 congruence on the complete computed table, not only selected successes.

Informative outcome. A recurrence or congruence mismatch rejects the coefficient pipeline. A useful research output is a surviving infinite residue family paired with a clearly missing argument, rather than a longer list of nonzero coefficients.

Bridge still needed. Unknown: exclude every possible prime and prime-power zero and then use multiplicativity for coprime factors. Finite congruence certification cannot perform those universal exclusions.

Established / proposed. The Hecke relations and Ramanujan congruence are classical. This exact exclusion inventory is proposed reproduction work without novelty or proof claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2603.29970

023 Lang-Trotter conjecture

Simulated team: arithmetic-statistics researcher

Target. For a fixed non-CM elliptic curve E/Q and integer r with positive Lang-Trotter constant C_E,r, predict #{p≤X good: a_p(E)=r}~C_E,r√X/log X; handle forced-zero traces separately.

Restricted question. Does a trace-counting and local-factor pipeline distinguish an algebraically impossible trace from a zero merely observed in a short sample?

Approach. Use E1:$y^2=x^3-x+1$ and E2:$y^2=x^3-2x+1$. Since E2 has rational two-torsion (1,0), its good odd-prime traces are even. The trace-one local factor must therefore be zero before any fitted prediction. Trace two supplies a comparison, while further local conditions must be checked before asserting a positive asymptotic constant.

Challenge. A sample containing no trace-one event does not itself prove impossibility. Fixed-modulus congruence frequencies also do not justify a matrix model whose resolution grows with p; the relevant Galois image cannot be replaced casually by the full matrix group.

Next task. Before new counts, specify E1 and E2 modular Galois-image models at m=2,3,6, documenting the method and any independence assumptions. Enumerate permitted trace residues and local factors, enforcing the E2 trace-one factor zero. Compare those models against the completed data only after the image computation is certified; growing-resolution tests remain unexecuted.

Baseline. Each curve has 166 included primes. E2 has zero odd traces, five trace-zero events, and six trace-two events. E1 has 54 odd traces but also zero observed trace-one events, illustrating the distinction.

Informative outcome. Any E2 odd trace or positive trace-one local factor rejects the pipeline. Next require the actual modular Galois-image model before increasing the resolution or fitting constants.

Bridge still needed. Unknown: quantitative joint control of congruences and archimedean trace size at growing resolution. The completed known-zero calibration supplies no sparse-event asymptotic.

Established / proposed. Two-torsion parity and point counting are established. Chen-Jones-Serban, Introduction equation (1.1), states the classical target; our completed control provides no new asymptotic estimate.

Proposed route. protocol design; readiness does not imply execution.

Completed result. New control executed through p<=1000. The second curve has 166 included good primes, all traces even, and zero trace-one events. The first curve also has zero trace-one events in this sample although 54 traces are odd. See the exact local obstruction and point-count output below.

https://arxiv.org/abs/2006.11269

024 Leopoldt’s conjecture

Simulated team: p-adic-number-theory researcher

Target. For every number field K and rational prime p, the p-adic logarithms of global units have rank r_1+r_2-1; equivalently the Leopoldt p-adic regulator is nonzero.

Restricted question. Can a certified precision ledger distinguish a nonzero regulator minor from a determinant whose apparent zero merely reflects insufficient p-adic precision?

Approach. Begin with totally real non-Galois cubic fields and a completely split odd prime. A rank-two unit basis gives a three-by-two logarithm matrix; test two-by-two minors rather than an undefined determinant of the rectangular matrix. Record the exact embeddings, unit-basis certificate, absolute input precision, and propagated output precision.

Challenge. Agreement at two nominal precisions is not certification unless all arithmetic losses are tracked. A small regulator valuation proves only the selected case, while a zero residue can reflect precision loss rather than regulator vanishing.

Next task. Order monic irreducible cubics with coefficients in -3,...,3 by positive nonsquare discriminant and then coefficients; select the first five distinct fields admitting a completely split odd prime below 101. Require certified field and unit computations. Define embedding choices and calculate minors at 50 and 100 absolute p-adic digits only after those prerequisites are met.

Baseline. For an output a+O(p^N), nonvanishing is certified only when v_p(a)<N. Change the unit basis by a unimodular integer matrix and verify that the resulting rank conclusion is unchanged.

Informative outcome. A discrepancy under a certified basis change rejects the implementation. High valuations earn a documented precision increase only if a bound predicts sufficient digits; do not call an unresolved minor a counterexample.

Bridge still needed. Unknown: a structural nonvanishing argument for every field and prime. Individual certificates or a pattern among small discriminants do not prove the universal rank assertion.

Established / proposed. p-adic regulators and unit computations are established. The cited mod-p variant is related context, not an equivalence with its stronger unit-regulator statement.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1805.00131

025 Lehmer’s Mahler measure problem

Simulated team: algebraic-number-theory researcher

Target. There is a constant $c>1$ such that every nonzero algebraic integer α that is not a root of unity has Mahler measure $M(\alpha)=\prod_{\sigma} \max(1,|\sigma(\alpha)|)\geq c$.

Restricted question. Among a fixed bounded polynomial family, which exact root configurations produce the smallest certified nontrivial Mahler measures, and do proposed resultant inequalities survive them?

Approach. Enumerate monic polynomials of degrees two through eight with remaining coefficients in $\{-1,0,1\}$. Remove powers of x and cyclotomic factors exactly, then factor over $\mathbb{Q}$ and deduplicate irreducible factors. Compare reciprocal and nonreciprocal factors while preserving degree; the conjecture concerns a degree-independent gap, not a minimum within this finite family.

Challenge. A degree-eight search cannot rule out small measures at higher degree. Floating-point roots near the unit circle are especially unreliable, and a proposed resultant bound may weaken with degree fast enough to miss the target completely.

Next task. Specify certified complex root isolation and Mahler-measure interval evaluation before running the enumeration. Compute all relevant pairwise resultants exactly for a chosen auxiliary family, which must be written down first. Rank the ten smallest intervals only when disjointness certifies their order; otherwise report ties as unresolved.

Baseline. Cyclotomic polynomials and powers of x must have measure one and be removed for the stated reasons. Include Lehmer’s degree-ten polynomial separately as a known reference outside the search box, not as an enumerated discovery.

Informative outcome. Any supposed universal lower bound contradicted by a certified example is rejected immediately. A useful output isolates the exact degree-dependent loss in the candidate inequality; reproducing familiar low measures is calibration only.

Bridge still needed. Unknown: an arithmetic inequality with a positive gap uniform in degree for all permitted algebraic integers. No finite coefficient box establishes that uniformity.

Established / proposed. Mahler measure, reciprocal root geometry, and auxiliary resultants are established. The proposed search audits an inequality that is still to be designed; novelty is not asserted.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1810.13363

026 Lehmer’s totient problem

Simulated team: multiplicative-number-theory researcher

Target. If a positive integer n satisfies φ(n)|(n-1), then n is prime or n=1; equivalently no composite n satisfies this divisibility.

Restricted question. Can exact rational interval bounds prune a finite squarefree-product search while preserving every candidate accepted by complete enumeration?

Approach. A hypothetical composite counterexample is odd and squarefree. Write n/φ(n)=∏_(p|n)p/(p-1), but test (n-1)/φ(n) exactly: replacing it by n/φ(n) discards the decisive 1/φ(n) correction. Bound this correction alongside the product on every branch, including its dependence on all remaining allowed factors.

Challenge. A gap between an approximate ratio and an integer proves nothing without enclosing the complete branch. Infinite searches also require a bound on the number and sizes of remaining prime factors, which this finite test does not provide.

Next task. Use subsets of {3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59}; there are 65536 subsets. Enumerate products exactly, separate n=1 and single-prime subsets from composites, and verify divisibility by integer remainders. Before pruning, specify rational lower and upper bounds for each unfinished branch.

Baseline. The complete finite enumeration is the oracle for auditing a proposed pruning rule. Every single-prime subset must pass the divisibility test, so accidentally discarding these controls reveals a normalization or integrality error.

Informative outcome. One incorrectly pruned solution invalidates the rule. A sound rule earns a report of saved branches and its exact hypotheses; if the savings rely on the sixteen-prime ceiling, label that limitation explicitly.

Bridge still needed. Unknown: extend certificates to arbitrarily many and arbitrarily large prime factors. Finite pruning efficiency does not establish that all infinite continuations terminate.

Established / proposed. Odd-squarefree restrictions and prime-ratio products are established. The proposed finite certificate audit is methodological and makes no novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2409.17076

027 Carmichael’s totient conjecture

Simulated team: constructive-number-theory researcher

Target. For every positive integer n, there is a distinct positive integer m with φ(m)=φ(n).

Restricted question. For each totient value v≤10000, can complete inverse-totient enumeration reveal which equal-totient pairs are explained by a fixed substitution vocabulary?

Approach. If φ(n)=v, each prime divisor p of n satisfies p-1|v, and each exponent e satisfies p^(e-1)(p-1)|v. Use those finite possibilities to recurse in increasing p, maintaining the exact remaining multiplicative factor. This enumerates all preimages of a bounded totient value without imposing an unjustified bound on n.

Challenge. Finding no partner below a cutoff on n would be meaningless because a partner may be larger. Even complete preimages for v≤10000 cannot establish that one finite substitution vocabulary covers every integer.

Next task. Design the recursion with separate handling of p=2 and v=1, where φ(1)=φ(2)=1. Fix the initial vocabulary to n↔2n for odd n and same-totient replacement of an explicitly enumerated prime-power block while preserving coprimality with the untouched block. Prove the recursion complete before generating coverage statistics.

Baseline. Check each generated n by independently evaluating φ(n), and compare small v with direct enumeration using the proven inverse candidates as a complete bound. Remove duplicate preimages produced by equivalent recursion branches.

Informative outcome. A missing known preimage rejects the recursion. A preimage class unexplained by the vocabulary motivates a specific new identity; it must not be called a unique-totient counterexample unless complete enumeration actually gives one.

Bridge still needed. Unknown: a partner construction covering every n, especially the families left after elementary doubling identities. A finite inverse table is evidence about the construction’s coverage only.

Established / proposed. Inverse-totient recursion and elementary equal-totient identities are established. This revised coverage study is a proposed organization of known tools without originality claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/math/9907204

028 Zaremba’s conjecture

Simulated team: continued-fractions researcher

Target. There exists an absolute integer A such that every $q \geq 2$ admits $1 \leq a < q$ with $\gcd(a,q)=1$ and a finite continued fraction $a/q=[0;b_1,\ldots,b_k]$ whose partial quotients all satisfy $b_i \leq A$.

Restricted question. For the stronger classical proposed bound $A=5$, can exact enumeration provide a witness for every $q \leq 2000$, and does a repair preserve both denominator and digit bound?

Approach. Generate digit words using matrices $M_b=[[b,1],[1,0]]$ for $b=1,\ldots,5$. Track numerators and denominators exactly and deduplicate rational values. Permit both finite continued-fraction representations when the final digit is one; a canonicalization convention must not silently reject a valid bounded-digit witness.

Challenge. Coverage modulo many integers does not imply coverage of individual denominators. A repair that increases the maximum digit or changes the desired denominator solves a different task, and a small exhaustive table may contain no genuine missing denominators to repair.

Next task. Enumerate all digit words whose denominator does not exceed 2000, recording exact fractions and a shortest witness for each q. Independently scan coprime a/q and test both finite representations. Write any repair identity symbolically before experimenting; if every denominator is covered, report calibration success without manufacturing a research obstruction.

Baseline. Cross-check matrix multiplication against the denominator recurrence $q_j=b_j q_{(j-1)}+q_{(j-2)}$, with $q_{-1}=0$ and $q_0=1$. Keep $q=1$ as a boundary state, excluding it from the target table.

Informative outcome. An alleged missing q with a witness in the independent scan rejects the enumeration. A useful repair must come with an exact invariant and work for a specified unbounded denominator family; isolated short words are insufficient.

Bridge still needed. Unknown: cover every denominator with some absolute bound A. Proving the stronger A=5 statement would suffice; a repair whose digit bound grows with q would not establish the original existential-A target.

Established / proposed. Continued-fraction semigroups and local-versus-global coverage are established. This witness audit is a calibration proposal, not a new coverage theorem.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2603.14116

029 Schanuel’s conjecture

Simulated team: transcendence-theory researcher

Target. For every n and Q-linearly independent $z_1,\dots,z_n \in \mathbb{C}$, the field $\mathbb{Q}(z_1,\dots,z_n,e^{z_1},\dots,e^{z_n})$ has transcendence degree at least n.

Restricted question. Can a dependency ledger distinguish proved finite nonvanishing checks for low-degree polynomials in e and π from the unknown uniform algebraic-independence statement?

Approach. Use all nonzero polynomials $P(X,Y)=c_0+c_1X+c_2Y+c_3X^2+c_4XY+c_5Y^2$ with $c_i \in \{-2,\dots,2\}$. Separate univariate cases covered by known transcendence of e or π from mixed cases. Interval exclusion of zero is a rigorous statement about one $P(e,\pi)$, not a symbolic independence theorem.

Challenge. Increasing coefficient or degree cutoffs cannot replace a uniform argument. Functional algebraic independence does not automatically survive specialization, and an interval containing zero is unresolved rather than evidence that a polynomial relation exists.

Next task. Normalize coefficient vectors by common gcd and overall sign to avoid duplicates. Evaluate with certified intervals at 128,256,512 bits, stopping at the fixed precision cap and labeling unresolved cases. First choose an interval package with outward rounding and write the dependency rules allowed in the symbolic ledger.

Baseline. Constants and univariate polynomials calibrate the evaluator; exact duplicate normalization is checked independently. Use Waldschmidt’s Conjecture 5.1 for the full target, keeping its hypotheses attached to every conditional implication.

Informative outcome. A certified nonzero interval resolves only that listed polynomial. A useful next research question identifies one unsupported specialization or logarithmic transformation with exact hypotheses; a long table of exclusions alone does not earn a higher-degree scan.

Bridge still needed. Unknown: a uniform algebraic-independence argument. Even proving full independence of e and π would address only a special consequence, not all complex tuples in Schanuel’s conjecture.

Established / proposed. Classical transcendence theorems and interval certification are established. This dependency audit is proposed expository and methodological work without novelty claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://webusers.imj-prg.fr/~michel.waldschmidt/articles/pdf/AWSLecture5.pdf

030 Four exponentials conjecture

Simulated team: logarithmic-forms researcher

Target. If $x_1,x_2\in\mathbb{C}$ are $\mathbb{Q}$-linearly independent and $y_1,y_2\in\mathbb{C}$ are $\mathbb{Q}$-linearly independent, at least one of the four numbers $\exp(x_i y_j)$ is transcendental.

Restricted question. For integers $2\le a,b,c,d\le 30$ with both pairs (a,b) and (a,c) multiplicatively independent, can exact dependency tests and interval arithmetic certify the logarithmic determinant is nonzero?

Approach. Set $D=\log(a)\log(d)-\log(b)\log(c)$, using real logarithms. If $D=0$, the logarithm matrix has rank one; independence of (a,b) and (a,c) supplies the two required rational-independence hypotheses. This explains precisely why a nonzero-determinant theorem would settle this rational-positive special case, rather than invoking a vague near-cancellation analogy.

Challenge. Bounds for linear forms in logarithms do not directly bound this quadratic determinant. Removing multiplicative dependencies is essential, since exact zero determinants are otherwise abundant and do not contradict the conjecture.

Next task. Factor a,b,c exactly and check ranks of their prime-exponent vectors for the two independence conditions. Evaluate D for every retained quadruple with certified intervals at 128,256,512 bits; keep undecided intervals explicitly. Choose the interval implementation and a precise proposed lower bound before an optimization search.

Baseline. The tuple $(2,4,3,9)$ gives $D=0$ but fails the first independence condition and must be removed. Apply row and column swaps to verify determinant signs and invariant nonvanishing decisions.

Informative outcome. A zero interval caused by an omitted dependence rejects the filter. Small certified determinants constrain any proposed quantitative bound; they are not candidates for counterexamples unless exact vanishing is independently proved.

Bridge still needed. Unknown: a lower-bound or nonvanishing argument for all admissible logarithm matrices, followed by extension beyond positive rational exponentials to the full complex statement.

Established / proposed. The logarithmic reformulation is classical; Waldschmidt’s Conjecture 5.11 states the full target. This finite audit is a proposed reproduction exercise without a new transcendence claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://webusers.imj-prg.fr/~michel.waldschmidt/articles/pdf/AWSLecture5.pdf

031 Irrationality of Euler’s constant

Simulated team: approximation-theory researcher

Target. Euler’s constant γ=lim_(n→∞)(Σ_(k=1)^n 1/k-log n) is irrational.

Restricted question. Does a selected known integral construction yield linear forms whose fully cleared denominators and additional logarithmic terms permit an irrationality criterion?

Approach. Demand an explicit decomposition L_n=A_nγ+B_n+Σ_j C_(n,j)log α_(n,j), with every coefficient and algebraic argument specified. Track the integral remainder together with the denominator-clearing multiplier D_n. Optimization is meaningful only after showing how the logarithmic terms are eliminated or rigorously controlled; a small displayed L_n is not sufficient.

Challenge. Denominator growth can erase every gain in approximation quality. An irrationality criterion cannot assume that the extra logarithmic combination is nonzero or irrational in a way that already requires information about γ.

Next task. During one bounded literature-design session, select an exact theorem or displayed construction from the cited Lagarias survey, transcribe its formulas and hypotheses, and identify the necessary denominator lemma. Only after that selection compute n=1,...,12; polynomial weights are not yet defined and no optimization run is ready.

Baseline. Use γ≈H_n-log n solely as a numerical evaluation control, with a proved error interval. The baseline must also demonstrate why ordinary harmonic approximations do not automatically give useful integer linear forms after denominator clearing.

Informative outcome. Deliver a complete coefficient-and-remainder ledger or a specific obstruction inequality showing that D_n|L_n| fails to decrease. If residual logarithms remain uncontrolled, stop and record that gap before trying more polynomial weights.

Bridge still needed. Unknown: construct nonzero integer linear forms A_nγ+B_n tending to zero, or another fully justified irrationality criterion whose auxiliary terms are controlled. The chosen integral representation alone does not provide this.

Established / proposed. Integral representations and irrationality criteria are established; the proposed improvement has no formula or novelty claim yet. This card is deliberately marked protocol design.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1303.1856

032 Normality of π

Simulated team: ergodic-theory researcher

Target. For every integer base $b \geq 2$ and every finite digit block w of length k, the limiting frequency of w in the base-b expansion of π is b^{-k}.

Restricted question. In the binary special case, can a precisely written BBP-related orbit recurrence yield a discrepancy inequality for the particular orbit of π across successive scales?

Approach. For binary digits d_i, define $C_w(N)=\#\{1 \leq i \leq N-k+1 : d_i \cdots d_{i+k-1}=w\}$ and $D_k(N)=\max_w |C_w(N)/(N-k+1)-2^{-k}|$. Separate a numerical test of this statistic from a theorem controlling the exact deterministic forcing in the BBP orbit. Almost-everywhere normality is not an orbit-specific estimate.

Challenge. A favorable finite digit sample cannot establish limiting frequencies, even for one fixed block length. Normality in base two would imply normality in its integer powers, but it does not automatically give the every-base target stated here.

Next task. First transcribe the precise orbit recurrence and its assumptions from the cited source. Then specify independently certified binary digits through 2^{20} and compute $D_k(2^j)$ for $4 \leq j \leq 20$ and $1 \leq k \leq \min(12,j)$. A proposed contraction inequality must be fixed before looking at those discrepancies.

Baseline. Use a periodic binary sequence as a nonnormal control and a seeded fair-bit sample as a statistical comparison only. Verify overlapping-block counting and endpoint normalization on short hand-checkable strings.

Informative outcome. One certified violation rejects a proposed universal contraction inequality. Passing this grid only motivates a proof attempt for that fixed recurrence; it must not be reported as proving normality or statistical independence of π's digits.

Bridge still needed. Unknown: an all-scale bound for this one orbit, every block length, and eventually every integer base. Each quantifier requires an argument beyond the finite binary experiment.

Established / proposed. BBP formulas and Lagarias's orbit formulation are established. This discrepancy audit is proposed work with no new normality criterion claimed.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/math/0101055

33-55 / Discrete mathematics

033 Hadwiger’s conjecture

Graph minors researcher (simulated)

Target. For every finite simple graph G and integer k≥1, χ(G)≥k should imply that G contains a K_k minor. The working restriction is k=7.

Restricted question. Which complete-minor certificates survive nonadjacent-vertex identification? Ordinary Hajós joins are a separate, elementary transfer benchmark, not the research obstruction.

Approach. Track seven disjoint connected branch sets through graph operations, but begin by separating two operations previously conflated. A Hajós join transfers a certificate whenever the auxiliary deleted edge is not a bridge. Identification of two nonadjacent vertices can instead merge branch sets and needs a genuinely different argument.

Challenge. The join lemma supplies no identification lemma. Restricting identifications to vertices already in one branch set would make the experiment easy while excluding the cases it was supposed to investigate.

Next task. Audit the accompanying elementary lemma; implement its contractions on successive joins with K7. Before any identification search, specify allowed identifications and the exact certificate transformations to be tested.

Baseline. Delete x_i y_i in disjoint G_i, identify x_1 with x_2, and add y_1 y_2. A surviving x_2-y_2 path replaces the missing first edge and recovers G_1 as a minor. The supplied note proves this with its non-bridge hypothesis.

Informative outcome. A failed certificate transfer on this baseline is an implementation error. For identification, an explicit example destroying a proposed transfer rule would reject that rule without rejecting Hadwiger.

Bridge still needed. A bridge would need a theorem covering unrestricted nonadjacent identification within an appropriate full construction calculus, with all seven branch sets preserved or rebuilt. That theorem is presently missing.

Established / proposed. The join definition is established and the transfer argument is elementary. This revision records a calibration lemma and a separate unresolved mechanism, claiming no mathematical novelty.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2609.06867

https://arxiv.org/pdf/1506.04225

034 Erdős-Hajnal conjecture

Extremal graph theorist (simulated)

Target. For every finite graph H, there should exist c(H)>0 such that every H-induced-subgraph-free graph on n vertices has a clique or independent set of size at least n^c(H).

Restricted question. In sampled connected P6-free graphs of clique number at most three, does either a dominating set of size three exist or an anticomplete pair with both sides at least n/4?

Approach. Replace an unspecified decomposition heuristic by that explicit, falsifiable dichotomy. Compute domination and the largest balanced anticomplete pair exactly, then compare failed dichotomy instances with their clique and independence numbers. The aim is to identify losses a recursive argument would actually have to accommodate.

Challenge. The dichotomy is only a proposed diagnostic and may be false. Even if every sample satisfies it, recursion could lose too many vertices to yield a positive exponent.

Next task. Using seed 34, propose at most 6,000 twelve-vertex random graphs at edge probabilities 0.2, 0.5, and 0.8; retain at most one hundred passing the stated filters.

Baseline. Use induced-subgraph enumeration to certify P6-freeness, subset enumeration for domination and independence, and three-way vertex assignments for disjoint anticomplete pairs. Complete graphs, empty graphs, and the five-cycle provide known checks of these individual routines.

Informative outcome. One exactly verified violating graph rejects the dichotomy. Preserve its adjacency matrix and optimal statistics; successful cases provide measurements only, with the sample's rejection rate reported.

Bridge still needed. A successful route would require a universally valid replacement dichotomy, a loss-controlled recurrence, and extension beyond the particular forbidden path. None of those implications follows from this finite comparison.

Established / proposed. The cited 2026 paper establishes additional special Erdős-Hajnal cases. Forbidden-subgraph decompositions are established; this bounded dichotomy test is exploratory and carries no originality claim.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2606.06258

035 Gyárfás-Sumner conjecture

Induced subgraph specialist (simulated)

Target. For every finite tree T and integer w, there should be a finite f(T,w) bounding χ(G) for all finite T-induced-subgraph-free graphs with clique number at most w.

Restricted question. Can failed rooted embeddings of a particular fifteen-vertex binary tree be converted into a precisely defined vertex partition whose color requirements are bounded?

Approach. Define the tree on vertices 0 through 14, joining each positive i to floor((i-1)/2). Record why extending a rooted induced embedding fails, including forbidden adjacencies to previously chosen vertices. The missing design step is a rule turning those records into a verifiable coloring partition.

Challenge. A greedy embedding failure is not evidence that the tree is absent. Different roots or embedding orders can remove the apparent obstruction, so an unrestricted exact search must arbitrate every failure.

Next task. Write the obstruction-record schema and partition rule first. Then use the second and third Mycielski iterates of C5, with 23 and 47 vertices, as explicitly generated test inputs.

Baseline. Construct M(G) with original vertices, duplicate vertices, and one apex; retain original edges, connect duplicates to original neighbors, and connect the apex to duplicates. Earlier smaller iterates are deliberate size-based negative controls for the fifteen-vertex embedding.

Informative outcome. An embedding found after a greedy failure invalidates that obstruction as a certificate of absence. A partition with uncontrolled interactions rejects the proposed coloring rule even when its individual parts look simple.

Bridge still needed. A proof would need a root-independent structural statement valid for every clique-bounded input and every finite tree, plus a recurrence producing finite coloring bounds. The current protocol supplies neither.

Established / proposed. Mycielski graphs, induced-tree embeddings, and χ-boundedness are established. This revision isolates an undefined implication as protocol design rather than presenting graph generation as a finished research method.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/html/2606.02030v1

036 Graph reconstruction conjecture

Graph reconstruction researcher (simulated)

Target. Every finite simple graph with at least three vertices should be determined up to isomorphism by the multiset of its unlabeled vertex-deleted induced subgraphs.

Restricted question. How often do separators of size at most two supply unambiguous card alignments, and exactly which attachment ambiguities survive those separator signatures?

Approach. In each card enumerate small separators and record component sizes, attachment neighborhoods, and multiplicities. Compare candidate alignments across the deck, preserving all surviving possibilities instead of making arbitrary symmetry choices. Return an ambiguity record whenever these constraints do not determine a unique reconstruction.

Challenge. Pairwise-compatible card alignments need not be globally compatible. A separator-free or highly symmetric card may provide no anchor; reporting such cases as unresolved is essential.

Next task. Run the proposed alignment procedure on the complete graph-atlas collection of connected graphs of orders three through seven; compare its outputs with an independent canonical-label deck lookup.

Baseline. Canonicalize each card by exhaustive vertex permutations at these small orders, retain repeated cards with their multiplicities, and hide the original graph during alignment. Paths, cycles, stars, and complete graphs check asymmetric, separator-rich, and symmetric behavior separately.

Informative outcome. A wrong reconstruction or discarded true alignment falsifies the procedure. An unresolved family is useful only if its exact competing attachments and the first missing global constraint are saved.

Bridge still needed. The full conjecture requires reconstruction even when no small separator exists, with no access to a finite atlas. An ambiguity-resolution theorem covering arbitrary graph structure is still needed.

Established / proposed. Reconstruction through separations is established, including the cited interval-graph result. This is an exact small-order calibration of a proposed alignment diagnostic, not a new reconstruction theorem.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2504.02353

037 Union-closed sets conjecture

Entropy and finite-set researcher (simulated)

Target. Every finite union-closed family F of finite sets containing a nonempty member should have an element occurring in at least |F|/2 members.

Restricted question. Are all 51 equality families in the completed four-element pilot exactly families of unions of disjoint nonempty blocks, after unused ground elements are removed?

Approach. Make the equality cases the immediate object of study. For each family identify its minimal nonempty members, test whether they partition its support, and compare the family with every union of those blocks. Separately record sections F0 and F1 obtained by removing a chosen element.

Challenge. A classification on four labeled elements is not a universal equality theorem. Section inequalities inferred from it could fail when additional elements introduce dependencies absent from the pilot.

Next task. Recover the four-element equality families, quotient by all 24 relabelings, and test the explicit block-union condition exactly; retain any violating family as a bitmask certificate.

Baseline. The prior pilot counted 4,958 eligible labeled families and 51 attaining maximum frequency one-half. Those are completed results. The new orbit classification and section analysis have not been executed.

Informative outcome. A counterexample rejects the proposed finite classification. Complete agreement gives a checked small-universe description worth comparing with established equality results before proposing a general lemma.

Bridge still needed. Any useful extension must prove a structural statement for arbitrary finite support, including families with unequal element frequencies. The existing count and a successful equality classification cannot establish that extension.

Established / proposed. Union closure, section identities, and entropy bounds are established; Gilmer's paper supplies direct context. The new classification is an unexecuted calibration task with no novelty claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result. Retained finite verification: all 4,958 eligible union-closed families on a labeled four-element ground set satisfy the half-frequency property; 51 attain equality. The general conjecture remains open in this notebook.

https://arxiv.org/abs/2211.09055

038 Caccetta-Häggkvist conjecture

Directed graph and SAT researcher (simulated)

Target. Every finite loopless simple digraph on n vertices with minimum out-degree at least r>0 should contain a directed cycle of length at most ceil(n/r).

Restricted question. Can exact small directed-triangle encodings expose short combinations of neighborhood constraints explaining infeasibility at minimum out-degree ceil(n/3)?

Approach. Introduce Boolean variables x_{ij} for directed edges, impose $x_{ij}+x_{ji}\leq 1$, and require $x_{ij}+x_{jk}+x_{ki}\leq 2$ for every ordered triple of distinct vertices. Add the stated minimum out-degree constraints. Inspect certified contradictions for reusable local counting patterns instead of interpreting solver runtime as evidence.

Challenge. A proof certificate at a fixed order need not produce an order-independent inequality. Restricting to oriented graphs is justified only because reciprocal edges already provide a short directed cycle.

Next task. Create the complete nine-vertex and twelve-vertex encodings, permit five minutes per solver run, and independently verify any unsatisfiability certificate; label a timeout unresolved.

Baseline. At each order n, the circulant with steps 1 through floor((n-1)/3) supplies a satisfiable below-threshold control: three permitted positive steps cannot sum to n. Direct cycle enumeration and out-degree counts must verify it.

Informative outcome. Acceptance of a control or rejection of an invalid certificate tests the implementation. A verified threshold contradiction is finite calibration; a satisfying graph requires independent edge and cycle verification.

Bridge still needed. To address the conjecture, extracted inequalities must hold at arbitrary order and then cover cycle lengths beyond three. Neither extension is provided by solving the two finite encodings.

Established / proposed. Triangle constraints, directed circulants, and SAT certificates are established. The proposed certificate analysis is a bounded study with an explicit falsifier, not a claimed new general bound.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/html/2604.13700v1

039 Barnette’s conjecture

Planar graph and matching researcher (simulated)

Target. Every finite cubic, bipartite, planar, three-vertex-connected graph should contain a Hamiltonian cycle.

Restricted question. On explicit even prisms, can a perfect matching be trapped against all alternating-cycle exchanges of length at most eight that reduce the number of complementary cycles?

Approach. For each perfect matching M, the remaining edges form a two-factor. Enumerate M-alternating cycles of length at most eight, flip membership along each, and count resulting two-factor components. The research diagnostic is a local-search trap despite a known Hamiltonian cycle, not Hamiltonicity of the prisms themselves.

Challenge. A short-exchange trap does not obstruct longer exchanges or another perfect matching. Prisms are especially structured, so a successful local rule there would say little about arbitrary Barnette graphs.

Next task. Use C10×K2, C12×K2, and C14×K2; enumerate their perfect matchings in lexicographic order, inspect the first one hundred per graph, and preserve every verified trap.

Baseline. These prisms have faces of length at least ten, meeting the earlier large-face intent, but are already Hamiltonian: traverse each rim with one edge removed and join the two resulting paths using matching end spokes.

Informative outcome. One trap refutes the proposed universal short-improvement rule on the benchmark. If none appears, report only the examined matchings and component changes, without generalizing beyond them.

Bridge still needed. A useful bridge would control arbitrarily long faces and non-prism geometry, proving that a bounded or explicitly manageable sequence of exchanges always reaches one cycle. That structural theorem is unknown here.

Established / proposed. Matching complements and alternating exchanges are established. The cited maximum-face-eight result is distinct from this solved prism calibration; no new Barnette case is claimed.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2508.03531

040 Lovász’s Hamiltonian-path conjecture

Algebraic graph theorist (simulated)

Target. Every finite connected vertex-transitive graph should have a path containing each vertex exactly once.

Restricted question. Can a Hamiltonian path be assembled using one contiguous visit per subgroup orbit in two explicit vertex-transitive benchmark graphs?

Approach. Replace vague quotient lifting by a constrained assembly problem. For each orbit enumerate endpoint pairs of spanning paths inside its induced subgraph. Enumerate orders of the orbits, then choose compatible endpoint pairs joined by actual graph edges. Failure means this one-visit restriction failed, not that Hamiltonicity failed.

Challenge. A global Hamiltonian path may need to leave and revisit an orbit. Orbit compression can also conceal disconnected fibers, so endpoint matching cannot repair every quotient obstruction.

Next task. Use the Petersen graph on two-element subsets of {0,1,2,3,4} with cyclic label rotation, and the three-cube with the bitwise-complement involution; enumerate all permitted assemblies.

Baseline. The Petersen graph is vertex-transitive but non-Cayley and has a Hamiltonian path. The cube has a Hamiltonian cycle; its chosen complement-orbit fibers are nonadjacent pairs, deliberately testing a failure of the contiguous-fiber restriction.

Informative outcome. A lifted path is checked vertex by vertex. Failure on the cube is an expected negative control revealing a real restriction of the method, not a counterexample to Lovász.

Bridge still needed. A general route must allow controlled re-entry into fibers or prove that some suitable subgroup avoids this obstruction. Existing long-cycle bounds do not supply that missing Hamiltonian-path lifting theorem.

Established / proposed. Orbit quotients and Hamiltonian-path searches are established. This revision provides explicit positive and negative controls for a restricted assembly mechanism, making no originality claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2606.09742

041 Rota’s basis conjecture

Matroid and linear algebra researcher (simulated)

Target. Given n bases of an n-dimensional vector space, order each as a row so that every column is a basis. The rank-n matroid statement is a generalization; this original target is its representable case.

Restricted question. In binary rank-four instances, how far can row-swap descent travel on a constant-defect plateau before finding an improvement?

Approach. Define defect D as the sum, over four columns, of four minus binary column rank. A move swaps two entries within one row. Build the finite state graph and measure the shortest escape to smaller D that never increases D, distinguishing plateau length from failure of the conjecture.

Challenge. Existence of a zero-defect arrangement does not imply a monotone path to it. Restricting moves or search depth can create traps even in a completely solved rank.

Next task. With seed 41, generate twenty collections of four ordered binary bases by rejection sampling nonzero four-bit columns. Enumerate row permutations, then compute plateau components and shortest exits; cap each instance at two minutes.

Baseline. Rota's conjecture is already proved for every rank-four matroid by Kirchweger, Scheucher, and Szeider. At most (4!)^4 = 331,776 row arrangements occur per labeled instance; complete enumeration supplies ground truth when the time cap permits.

Informative outcome. A plateau without a descending exit refutes monotone descent from that state. A shortest escape exceeding four swaps rejects a four-move heuristic. Timeouts are unresolved, while a complete absence of zero-defect states signals an encoding error.

Bridge still needed. The conjecture needs an argument at arbitrary rank. Useful progress would explain why larger temporary defect or richer exchanges suffice uniformly, rather than extrapolating rank-four success. No such exchange theorem is presently supplied.

Established / proposed. Basis exchange and defect search are established methods. The contribution proposed here is a precisely measured benchmark of exchange geometry within a solved case, without a novelty claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SAT.2022.4

042 Hadwiger-Nelson problem

Geometric graph and constraint researcher (simulated)

Target. Determine the least number of colors needed to color the Euclidean plane so that points exactly one unit apart receive different colors.

Restricted question. Can two documented finite unit-distance modules be glued by an exact isometry to produce a graph with no five-coloring?

Approach. Begin with an input manifest, not a color-search claim. De Grey's published finite graph with no four-coloring is a possible module source, but the actual coordinate file, number field, and five-coloring certificate have not been selected. Only certified module data should enter the proposed gluing experiment.

Challenge. An abstract graph coloring certificate does not establish a unit-distance realization. Nor does absence of four-colorings establish five-colorability. Unverified coordinate approximations can silently create or delete edges.

Next task. Locate primary exact coordinate data and record its URL, digest, algebraic representation, edge equality predicate, and independently checkable four-color UNSAT and five-color SAT certificates. Stop at input selection if any requirement remains missing.

Baseline. First validate geometry and coloring on the equilateral triangle with coordinates (0,0), (1,0), and (1/2,√3/2): its unit edges admit three colors and exclude two. The cited de Grey paper supplies mathematical context, not an already imported coordinate dataset.

Informative outcome. After inputs are certified, identify oriented unit edges between two module copies, considering both planar isometries. A five-coloring rejects that candidate. Certified five-color impossibility would require separate exact-geometry review before supporting a six-color lower bound.

Bridge still needed. Even exhaustive search over a chosen finite module family would leave other constructions and the exact chromatic number unresolved. The immediate deliverable is a reproducible input specification; the proposed search remains conditional on completing it.

Established / proposed. Finite unit-distance graphs and SAT coloring are established. This revision removes an unsupported input assumption and proposes a conditional experiment, without claiming a new construction.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1804.02385

043 Planar unit-distance problem: exact growth

Discrete geometry and optimization researcher (simulated)

Target. Determine the asymptotic growth of u(n), the maximum number of unordered pairs at distance one among n distinct points in the Euclidean plane.

Restricted question. Can one explicit instance of the cited number-field construction be generated within a 200-point budget, and does pruning change its most frequent distance?

Approach. Separate construction feasibility from optimization. For each subset A compute both U(A), counting a fixed original distance, and R(A), the maximum multiplicity over all positive distances. Only after selecting the maximizing distance may the configuration be rescaled to unit distance; optimizing U alone answers a narrower question.

Challenge. An asymptotic number-field tower does not automatically provide usable small coordinates. A toy grid cannot validate that construction, and improving one finite configuration cannot establish an asymptotic exponent.

Next task. First specify the number field, integral basis, planar embedding, coefficient box, exact point list, and distance-comparison procedure. If no instance with at most 200 points is established, report that no suitable input was established within this design pass; keep feasibility unresolved.

Baseline. An explicitly separate toy control is $\{0,\ldots,8\}^2$. Prune its 81 points to 50 by maximizing R after each deletion, breaking ties lexicographically; compare fixed-unit-distance pruning and 100 random 50-point subsets with seed 43. These controls remain unexecuted.

Informative outcome. Different maximizers for U and R expose the original statistic mismatch. A complete exact-coordinate manifest makes a genuine construction experiment possible. Neither toy success nor failure measures the asymptotic construction's strength.

Bridge still needed. The full problem needs bounds uniform as n grows, with all construction parameters tracked. The former n^(1+o(1)) conjecture has been disproved by the cited 2026 work; this target asks for the remaining exact growth and does not reinstate it.

Established / proposed. Repeated-distance constructions and rescaling are established. The proposed feasibility audit and corrected pruning statistic claim no novelty.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2607.05374

044 Hadwiger-Boltyanski illumination conjecture

Convex geometry researcher (simulated)

Target. Every convex body with nonempty interior in Euclidean dimension d should be illuminable by at most 2^d directions.

Restricted question. Do eight directions from a prescribed small integer set illuminate each member of a reproducible collection of three-dimensional rational polytopes?

Approach. Compute an exact facet description and treat illumination as set cover. A direction u illuminates a vertex v precisely when $a_f \cdot u < 0$ for every outward facet normal incident to v. This yields a Boolean coverage matrix whose eight-column covers can be checked without floating-point tolerances.

Challenge. A finite direction dictionary is incomplete. Failure to cover a polytope with that dictionary cannot refute illumination, even if the discrete set-cover instance is solved exhaustively.

Next task. With seed 44, generate fifty full-dimensional hulls from twenty-eight distinct integer points in $[-5,5]^3$, allowing one hundred proposals. Try directions in $[-2,2]^3$ except zero, then $[-3,3]^3$; cap each set-cover solve at sixty seconds.

Baseline. The cube $[-1,1]^3$ is illuminated by its eight inward body-diagonal directions. Its corner sign constraints also require eight separate direction orthants. Check this exact positive and necessity benchmark before processing sampled hulls.

Informative outcome. A proposed cover is useful only after independently verifying every strict facet inequality. A missing eight-direction cover reports dictionary insufficiency or a timeout, not a counterexample. Repeatedly needed directions outside the first dictionary indicate how discretization distorts the search.

Bridge still needed. Passing finitely many polytopes does not control every convex body or dimension. A bridge would require a geometrically complete direction-selection theorem and a justified passage from polytopes to arbitrary bodies, including stability of strict illumination inequalities.

Established / proposed. Illumination criteria and set-cover formulations are established. The proposal supplies exact input generation, a finite dictionary, and a falsifiable calibration protocol, without claiming an original upper bound.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2501.10570

045 Lonely runner conjecture

Diophantine approximation researcher (simulated)

Target. For any k distinct positive integer speeds $v_1,\ldots,v_k$, some real time t satisfies $||t\,v_i|| \geq 1/(k+1)$ for every i, where $||\cdot||$ is distance to the nearest integer.

Restricted question. For fourteen speeds, do near-tight feasible time sets correlate with specified short additive relations among the speeds?

Approach. Use exact interval intersections. For a speed v and threshold δ, its permitted times in [0,1] are the closed intervals $[(m+\delta)/v,(m+1-\delta)/v]$, for $0 \leq m < v$. Intersect these lists with rational arithmetic at three fixed thresholds instead of claiming an uncomputed exact optimal margin.

Challenge. A relation-frequency correlation is not a structural reduction. Sampling bounded integer speeds can miss the arithmetic configurations that make a general proof difficult.

Next task. Using seed 45, retain five hundred gcd-one sets of fourteen distinct speeds in {1,...,100}, with two thousand proposals maximum. Test δ = 1/15, 1/15 + 1/1500, and 1/14; record relations $v_i+v_j=v_k$ and $v_i+v_j=v_k+v_l$ with distinct indices.

Baseline. The speeds {1,...,14} have optimal minimum separation 1/15; t=1/15 supplies a witness. Reproduce their exact interval intersection before the sample, checking closed endpoints so equality cases are not accidentally discarded.

Informative outcome. An empty intersection at 1/15 demands independent exact verification and would be mathematically significant. Emptiness at either larger threshold simply measures tightness. Any observed correlation must be compared with a separately generated holdout sample before being used as a prediction.

Bridge still needed. The conjecture quantifies over unbounded speed sets. A useful bridge must prove that arithmetic relations force a favorable time or reduce to a controlled lower-dimensional case; finite interval calculations provide examples and obstructions, not that theorem.

Established / proposed. Interval methods and additive relations are established. The suggested fixed-threshold diagnostic makes no novelty claim.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2609.06259

046 1/3-2/3 conjecture

Order theory and exact enumeration researcher (simulated)

Target. Every finite partially ordered set that is not a total order should have an incomparable pair (x,y) with 1/3 ≤ Pr(x precedes y) ≤ 2/3 in a uniformly random linear extension.

Restricted question. Can a specified local signature reliably identify balanced pairs, or do equal signatures conceal different extension probabilities?

Approach. Assign each ordered incomparable pair the signature consisting of its two incomparability degrees, number of common incomparable elements, and sizes of its two strict lower and two strict upper sets. Search for equal signatures containing both balanced and unbalanced pairs; such collisions directly test the proposed local explanation.

Challenge. Linear-extension probabilities depend on global constraints. Even a signature that separates all tiny examples may fail when distant order relations change the number of compatible extensions.

Next task. Reuse the five-element pilot, then sample one hundred posets on eight to ten elements with seed 46 from forward-edge DAGs of edge probability 1/3 followed by transitive closure; exclude total orders and allow one thousand proposals.

Baseline. Compute extension counts by dynamic programming over order ideals. For incomparable x,y, compare the count after adding x<y with the original count. Verify this procedure against permutation enumeration on the existing small pilot before evaluating new signatures.

Informative outcome. One balanced/unbalanced collision rejects the signature as a complete criterion. A collision-free sample warrants an independent holdout, not a universal claim. Report exact rational probabilities and the two witness orders whenever a collision occurs.

Bridge still needed. A general proof needs a global counting inequality or a strengthened structural criterion valid for every finite poset. The recorded completed pilot proves only its finite cases; the signature analysis and larger sampling have not been executed.

Established / proposed. Ideal dynamic programming and local-poset statistics are established; their proposed comparison carries no novelty claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result. Retained finite verification: all 4,111 non-total labeled partial orders on five elements have a pair with ordering probability in [1/3,2/3]. No unrestricted result is inferred.

https://arxiv.org/abs/1610.00809

047 Erdős-Szekeres convex-polygon conjecture

Order-type and proof-certificate researcher (simulated)

Target. For every n≥3, any set of 2^(n-2)+1 planar points with no three collinear should contain n points in convex position.

Restricted question. Can a sound orientation encoding exclude a seven-gon-free 33-point configuration with either of two explicitly chosen convex-layer profiles?

Approach. Use triple-orientation variables and necessary rank-three consistency constraints. Study layer profiles (6,6,6,6,6,3) and (6,6,6,6,5,4), both totaling 33. The encoding must prove that every realizable configuration with the selected profile survives its constraints; abstract consistency alone does not guarantee geometric realizability.

Challenge. A satisfiable abstract orientation system may have no realization by points. Conversely, an overly restrictive layer encoding can create meaningless UNSAT results by deleting legitimate configurations.

Next task. Write the orientation, layer-membership, and forbidden-seven-gon constraints as a reviewable specification before solving. Validate the four-point target first; only then give each 33-point profile a two-minute trial, treating timeouts as unknown.

Baseline. The known four-point threshold is five points. The encoding should exclude five points without a convex quadrilateral, while accepting the triangle (0,0),(3,0),(0,3) with interior point (1,1), which has no convex quadrilateral.

Informative outcome. A failed small control invalidates the encoding. Checked UNSAT plus a proof of encoding completeness would exclude only the selected layer profile. A SAT result remains an abstract candidate until coordinates or a valid realization certificate are supplied.

Bridge still needed. Two profiles cover only a small part of the seven-gon question. The full conjecture requires all possible profiles and all n, or a uniform inductive principle; neither follows from this restricted search.

Established / proposed. Order types, convex layers, and SAT are established tools. The suggested profiles are test inputs, not a claimed new reduction. Constraint design remains unfinished and is explicitly the next deliverable.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2512.24061

048 Conway’s thrackle conjecture

Topological graph theorist (simulated)

Target. A finite simple graph drawn with Jordan-arc edges, where adjacent edges meet only at their common endpoint and nonadjacent edges cross exactly once transversely, should have at most as many edges as vertices.

Restricted question. Can a complete drawing-constraint encoding rule out small subdivided theta and dumbbell graphs as thrackles?

Approach. Keep every degree-two vertex and every edge while encoding crossing order and local rotation. Start with theta graphs whose three internally disjoint paths have lengths at least two and total length at most ten, plus two cycles joined by a path with at most ten total edges.

Challenge. Parity conditions alone need not guarantee a valid drawing. Suppressing a subdivision can change which edge pairs must meet, so an unproved reduction to a tiny graph is unsafe.

Next task. Specify edge-route contiguity, crossing rotations, and the global embedding test before launching searches. Validate the control drawings, then examine the first two cases in lexicographic path-length order with a five-minute cap each.

Baseline. A triangle and the five-cycle drawn as a pentagram are valid controls. A four-cycle cannot be a thrackle. The test suite must distinguish these cases using the complete drawing model, not only its parity relaxation.

Informative outcome. Acceptance of a four-cycle drawing invalidates the encoding or certificate. Rejection of one small theta graph concerns that exact subdivision. An abstract crossing schedule remains provisional until its planarized rotation system gives the required actual drawing.

Bridge still needed. A proof needs either unrestricted drawing constraints with a general obstruction theorem or a justified reduction of every minimal counterexample to a manageable family. The chosen edge budget supplies neither reduction, and no finite exhaustion of all possible subdivisions is claimed.

Established / proposed. Crossing-order encodings and small forbidden configurations are established approaches. This revision removes an unjustified suppression step and makes the missing encoding obligations explicit; it claims no new forbidden-family theorem.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2506.11808

049 Hadamard matrix conjecture

Combinatorial design researcher (simulated)

Target. For every positive integer k, there exists a 4k-by-4k matrix H with entries ±1 satisfying $HH^T = 4k\,I$.

Restricted question. Does appending two signs to each member of a complementary length-three binary quadruple always permit a complementary length-five quadruple?

Approach. For a sign sequence s of length n, define $C_s(t)=\Sigma_j s_j s_{(j+t \bmod n)}$. A quadruple is complementary when $\Sigma_i C_{(s_i)}(t)=0$ for every $1\le t<n$. The extension now means exactly $s_i'=(s_i,\varepsilon_{(i,1)},\varepsilon_{(i,2)})$, with eight independently chosen signs; original entries and their order remain fixed.

Challenge. This rigid append rule may fail even though Hadamard matrices of both corresponding orders exist. Failure would disprove the local lifting mechanism, not Hadamard existence or the broader Goethals-Seidel construction.

Next task. Enumerate all 256 quadruples of length-three sequences normalized to first entry +1, retain those satisfying the correlation equations, and test all 256 append choices for each. Store a successful extension or a complete failure certificate for every valid parent.

Baseline. The quadruple A=(1,1,1), B=C=D=(1,1,-1) is complementary: the constant sequence contributes 3 and each other sequence contributes -1 at either nonzero shift. The established Goethals-Seidel array converts suitable complementary circulants into a Hadamard matrix.

Informative outcome. One valid parent with no permitted extension refutes universal appendability in this explicitly defined experiment. A successful extension supplies only a small construction. If every parent succeeds, the result remains a finite calibration, not an induction theorem.

Bridge still needed. A route to all admissible orders would require a proved extension operation at arbitrary lengths and sufficient initial cases, or a different complete construction. This restricted family and its two-coordinate append rule are not known to supply either.

Established / proposed. Periodic complementarity and Goethals-Seidel arrays are established. The proposed append experiment is now defined mathematically; no originality or execution is claimed.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2306.16812

050 Turán’s tetrahedron problem

Hypergraph extremal researcher (simulated)

Target. The limiting maximum edge density of a three-uniform hypergraph containing no four vertices with all four triples should equal 5/9.

Restricted question. Which five-vertex induced statistics distinguish the balanced three-part construction from competing small tetrahedron-free examples, and what additional consistency constraints would a density certificate require?

Approach. Enumerate five-vertex triple systems and collect exact induced-type frequencies before proposing any inequality. A usable upper bound must combine these statistics through a universally valid averaging or positive-semidefinite certificate. No such inequality has yet been specified; this stage designs its inputs rather than claiming a local-density proof.

Challenge. An inequality fitted to a few constructions can fail on another hypergraph. Five-vertex counts alone may be too weak, and finite extremal densities need not lie below the asymptotic target.

Next task. Enumerate all 2^10 labeled five-vertex triple systems, retaining those with no tetrahedron, and classify their induced types and edge counts. Within a one-hour design budget, state any proposed certificate constraints explicitly or stop with the verified enumeration table.

Baseline. Use the cyclic three-part construction: include triples meeting all parts and triples with two vertices in part i and one in part i+1, indices modulo three. It has 54 edges at nine vertices and 136 at twelve, so comparisons require finite-size corrections.

Informative outcome. A single admissible hypergraph violating a proposed universal inequality rejects it. An enumeration table is calibration data, not an upper bound. An eventual certificate must include exact coefficients, a positivity check, and a derivation of its finite error term.

Bridge still needed. The target is ordinary Turán density. The recently solved uniformly dense variant with threshold 1/2 imposes an additional hypothesis and does not settle this 5/9 problem. A bridge requires an inequality valid for every tetrahedron-free hypergraph as its size grows.

Established / proposed. Induced-density and flag-algebra methods are established. The suggested statistic audit does not claim a new inequality.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2609.08336

051 Ryser’s hypergraph conjecture

Hypergraph matching researcher (simulated)

Target. Every r-partite r-uniform hypergraph, for r≥2, should have vertex-cover number τ at most (r-1) times its matching number ν.

Restricted question. Can one-edge perturbations of an explicit four-partite example with ν=2 and τ=6 reveal a locally different obstruction to small covers?

Approach. Construct two disjoint copies of a truncated affine-plane example. In copy c∈{0,1}, take edges (3c+x,3c+y,3c+(x+y mod 3),3c+(x+2y mod 3)) for x,y∈{0,1,2}; coordinate positions identify the four parts. Add one unused vertex labeled 6 to each part before perturbing.

Challenge. A tightly structured seed may have a very small feasible neighborhood. Inability to increase its cover number says little about arbitrary hypergraphs, while losing the matching constraint would invalidate the intended comparison.

Next task. With seed 51, make one hundred proposals deleting one existing edge and adding one absent tuple in $\{0,\ldots,6\}^4$. Retain only matching-number-two cases. Compute cover numbers by exact branch search with sixty seconds per proposal, marking unfinished calculations unknown.

Baseline. Each nine-edge copy is intersecting and has cover number three: any two vertices cover at most six edges, while a whole three-vertex part covers all nine. Two disjoint copies therefore have ν=2 and τ=6. Verify these exact controls independently.

Informative outcome. A retained instance with certified τ≥7 would contradict the r=4 inequality and require independent matching and cover certificates. More modest outcomes can identify why permissible local changes fail to destroy six-covers; they do not establish a global extremal classification.

Bridge still needed. The conjecture ranges over all r and matching numbers. A useful bridge would turn the observed cover structure into a provable decomposition or exchange principle. The stronger Lovász formulation has been disproved, so it must not be imported as an assumption.

Established / proposed. Truncated-plane examples and exact covering searches are established; this explicit neighborhood experiment makes no novelty claim.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2505.05339

052 Ryser’s Latin-square transversal conjecture

Latin square and matching researcher (simulated)

Target. Every Latin square of odd order should contain a transversal: one cell from each row and column with all chosen symbols distinct.

Restricted question. In cyclic Latin squares of orders five and seven, can every partial transversal of size n-1 reach an augmentable one by single-cell exchanges, and how many exchanges are needed?

Approach. Fix L(i,j)=i+j modulo n. A state is a valid partial transversal with n-1 cells. A move removes one selected cell and inserts one other cell while preserving size and all three distinctness conditions. Goal states admit a direct insertion completing a transversal; breadth-first search measures exact exchange distance.

Challenge. Cyclic squares already have known transversals and unusually strong symmetry. A disconnected exchange graph would refute this move rule's sufficiency, not the conjecture; successful distances need not persist in other Latin squares.

Next task. Enumerate states by their omitted row, omitted column, and bijection between remaining rows and columns, retaining those with distinct symbols. Run reverse breadth-first search from all goal states for n=5 and n=7; record unreachable components and maximum finite distance.

Baseline. For odd n, the diagonal cells (i,i) form a transversal because 2i runs through all residues. At n=7 there are at most $7^2 \cdot 6! = 35{,}280$ candidate states before symbol filtering, giving an explicit finite enumeration budget.

Informative outcome. A distance exceeding four rejects a four-exchange heuristic on the benchmark. An unreachable component disproves augmentation by these moves from every state. Verify each reported shortest path cell by cell and retain a representative of every unreachable component.

Bridge still needed. The experiment changes only the selected cells, never the Latin square. A general proof would need an exchange theorem or a different augmentation mechanism for arbitrary odd-order squares, with control independent of cyclic symmetry.

Established / proposed. Partial-transversal exchanges and cyclic-square transversals are established. This proposal defines a finite test of move connectivity, without claiming a new existence result.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1801.02893

053 Tutte’s 5-flow conjecture

Integer flow researcher (simulated)

Target. Every finite bridgeless graph should admit a nowhere-zero integer flow whose absolute edge values belong to {1,2,3,4}.

Restricted question. Do fundamental-cycle coordinate moves become trapped for a specified defect potential on four small bridgeless benchmark graphs?

Approach. Fix an orientation and a breadth-first spanning tree using lexicographic vertex order. Write each circulation as an integer combination of the resulting signed fundamental cycles. On coefficients $z \in \{-2,\dots,2\}^{\beta}$, define P as the number of zero-flow edges plus $\Sigma_e \max(0,|f_e|-4)$; a move changes one coefficient by one within the box.

Challenge. The chosen basis and coefficient box can create artificial traps. A valid five-flow may require coefficients outside this box even though it exists on the underlying graph.

Next task. Use the Petersen graph, three-cube, pentagonal prism, and hexagonal prism. Enumerate the coefficient boxes, connect permitted moves, and find components from which no non-increasing-P path reaches P=0; retain a witness state and its minimum escape barrier.

Baseline. These are known small flow-feasible graphs. Independently solve the signed flow equations with $f_e \in \{\pm1,\dots,\pm4\}$ and verify conservation before interpreting search failures. Their cycle ranks are at most seven, so each coefficient box has at most $5^7 = 78{,}125$ states.

Informative outcome. A positive barrier refutes unrestricted monotone descent for that basis and box. If the box contains no zero-defect state while the independent solver finds a flow, the obstruction is representational. Neither outcome challenges the conjecture on these already tractable graphs.

Bridge still needed. A proof would need a basis-independent escape principle, controlled enlargement of the coefficient range, or a structural reduction covering all bridgeless graphs. Finite landscape statistics alone do not bound barriers as cycle rank grows.

Established / proposed. Cycle-space coordinates and integer-flow optimization are established. The proposed potential and explicit state-space measurements are research diagnostics, without a novelty claim.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2601.12036

054 Conway’s 99-graph problem

Strongly regular graph researcher (simulated)

Target. Determine whether a simple graph on 99 vertices exists with degree 14, one common neighbor for each adjacent pair, and two common neighbors for each nonadjacent pair.

Restricted question. Can a complete rooted adjacency encoding produce a checked candidate or a checked impossibility certificate within an explicitly bounded first solve?

Approach. Fix a root whose fourteen neighbors form seven disjoint pairs. Label each of the other 84 vertices by the nonmatched pair of root-neighbors adjacent to it. Keep every adjacency among these 84 vertices as an independent variable, then impose the full matrix identity $A^2+A-12I=2J$ with symmetric zero-diagonal A and degree fourteen.

Challenge. Root labeling is legitimate only after proving that every graph with the required parameters admits it. Additional symmetry assumptions among the 84 vertices would restrict the problem and could make an UNSAT result misleading.

Next task. First encode the analogous rooted 3-by-3 rook graph. After its certificate passes, run the complete 99-vertex model for five minutes, storing either a candidate matrix, a checkable proof trace, or the explicit timeout status.

Baseline. The nine-vertex rook graph has parameters (9,4,1,2) and satisfies $A^2+A-2I=2J$. Its root has two matched neighbor pairs and four remaining vertices, providing a small positive control for the same labeling and common-neighbor equations.

Informative outcome. Any candidate must be checked directly against every degree and pairwise common-neighbor requirement. A failed rook control rejects the implementation. UNSAT is decisive only with a verified proof and a reviewed completeness argument; timeout gives no existence evidence.

Bridge still needed. Unlike an asymptotic conjecture, this finite target can in principle be settled by a complete certified encoding. The remaining bridge is proof of the rooted parametrization and encoding equivalence, followed by a certificate that survives independent checking.

Established / proposed. Strongly regular graph equations and rooted parameterizations are established. This proposal supplies a controlled exact search design, without claiming a new reduction or executed result.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2308.02978

055 Tutte’s 3-flow conjecture

Orientation and network flow researcher (simulated)

Target. Every finite 4-edge-connected graph should have a nowhere-zero integer flow with absolute edge values one or two.

Restricted question. Can prescribed-outdegree orientation and an exact integral lift recover three-flows on four explicit five-regular circulant graphs?

Approach. At a five-valent vertex, an orientation balanced modulo three has outdegree one or four. Choose half the vertices for each type and solve the prescribed-outdegree problem by assigning each undirected edge one tail. For an accepted orientation, solve $Bz=B1/3$ with $z_e\in\{0,1\}$; then $f=1-3z$ is an integer circulation with values 1 or -2.

Challenge. A modulo-three orientation cannot simply be treated as an ordinary integer flow. The integral lifting step is essential, and restricting to degree five leaves much of the full conjecture untouched.

Next task. For $n=6,8,10,12$, use the circulant with steps ±1, ±2, and $n/2$. Verify 4-edge-connectivity, enumerate every balanced type assignment, solve each edge-to-tail network flow, then lift any successful orientation and check conservation exactly.

Baseline. For K6, divide vertices into two triples, orient all cross edges from the high-outdegree triple to the low-outdegree triple, and orient each triple cyclically. This yields outdegrees four and one. The lift is feasible integrally because $z=1/3$ is fractionally feasible and network matrices are totally unimodular.

Informative outcome. Every successful signed flow receives a direct edge-bound and conservation check. Exhaustive failure across all type assignments would require independent verification of connectivity and model equivalence before suggesting a counterexample. A timeout or a failed single assignment has no such meaning.

Bridge still needed. A general proof must handle arbitrary degrees and graph structure or justify a reduction to a class such as five-regular graphs. These four already manageable examples calibrate orientation and lifting machinery; they supply no such reduction.

Established / proposed. Modulo-three orientations, prescribed-degree flows, and integral circulation lifting are established. The experiment specifies their composition without claiming novelty.

Proposed route. research protocol ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2609.08377

56-80 / Geometry, topology and algebra

056 Smooth four-dimensional Poincaré conjecture

Simulated four-manifold topologist

Target. Every closed smooth four-manifold homotopy equivalent to S4 is diffeomorphic to the standard S4.

Restricted question. Can the proof of Cha-Kim Theorem A be represented as a dependency-checked certificate, including its five-dimensional handlebody comparison and framing argument?

Approach. Replace the assumed four-dimensional move replay with extraction of a structured proof certificate from Cha-Kim's actual five-dimensional comparison. Record mapping-class identifications, attaching circles, framings, and the boundary Gluck-twist argument. Only after defining what a checker can verify should heuristic searches or non-geometric extensions be considered.

Challenge. The source deliberately uses a different method from diagram simplification. A dependency graph is not a formal proof, and varying a geometric fibered-knot parameter remains inside the solved family.

Next task. Allocate one design pass to Section 3: list objects, identifications, framings, and invoked lemmas. Stop before automated replay unless every certificate operation has a defined verification rule.

Baseline. Theorem A already standardizes every geometric, fibered-knot Calegari sphere. Section 3 supplies a proof to analyze, not the assumed replayable four-dimensional move list.

Informative outcome. A useful outcome is an explicit dependency graph whose framing and Gluck-twist assertions each have a cited justification. Missing representational details remain unresolved certificate obligations. Every invoked theorem remains a dependency unless its proof is independently formalized.

Bridge still needed. Certificate extraction only calibrates the workflow. Extending the proof to admissible non-geometric automorphisms requires replacing its geometric input; covering arbitrary homotopy spheres is a further unknown step.

Established / proposed. Theorem A and Section 3 are established input; structured certificate extraction is proposed design work. No published slide list, new case, or originality claim is assumed.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2411.10051

057 Smooth four-dimensional Schoenflies conjecture

Simulated specialist in smooth embeddings

Target. For every smooth embedding S3 into S4, both closures of the complementary components are diffeomorphic to the standard four-ball B4.

Restricted question. Can a chosen embedding be encoded with both complementary regions and their boundary identification, so every proposed simplification preserves the same embedding data?

Approach. Focus on three-sphere embeddings produced by a specified fibered-knot construction. Track both complementary four-manifolds simultaneously, searching for coordinated handle simplifications rather than simplifying each side independently. Use the shared boundary identification as additional data, hoping to expose cancellation opportunities hidden in either isolated description.

Challenge. Contractibility and a three-sphere boundary do not establish that a complementary region is a standard smooth ball. Coordinating two handle descriptions could increase complexity without adding an obstruction.

Next task. Specify the equatorial embedding's two handle descriptions and boundary-map convention, then define one paired cancellation and its checker. Park the search until this design has no unchecked identifications. Record orientation conventions explicitly for both copies of the common boundary.

Baseline. Begin with the equatorial S3 in standard S4, whose two complements are standard balls. Gabai's pseudo-isotopy approach is contextual groundwork, not a ready paired-move implementation.

Informative outcome. A boundary-map mismatch or a move changing the represented complement rejects the encoding. Successful equatorial verification establishes only a reliable representation and regression test.

Bridge still needed. A universal conclusion requires a valid standardization argument for arbitrary embeddings. Contractibility, matching homology, or finitely many successful paired descriptions cannot provide that bridge.

Established / proposed. Handle and pseudo-isotopy approaches are established; synchronized certificate searches across both complementary regions are the suggested experiment, not a novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2212.02004

058 Andrews-Curtis conjecture

Simulated computational group theorist

Target. Every finite balanced presentation of the trivial group can be reduced to the standard presentation using elementary Andrews-Curtis moves, without adding generator-relator pairs.

Restricted question. Does bidirectional search find shorter certificates between the exact published length-fourteen presentations, under a fixed elementary-move alphabet and node budget?

Approach. Use the published length-fourteen presentation frontier as a finite laboratory. Search bidirectionally between hard presentations and known equivalence classes, allowing temporary relator growth. Train move priorities on replayable successful sequences, while keeping the final certificate entirely in elementary Andrews-Curtis moves and separating stable from unstable moves.

Challenge. A finite search cannot disprove the conjecture, and successful equivalence to another unresolved presentation does not trivialize either one. Allowing stabilization would silently change the target problem.

Next task. Select one source certificate by its exact endpoints, define moves and canonicalization, and cap both search methods at 100,000 expanded nodes; check every returned elementary step independently.

Baseline. Replay a named machine-checkable certificate from the source and compare with unweighted breadth-first search on the same presentation pair. Existing certificates are calibration, not newly solved cases.

Informative outcome. Report verified path length, expanded nodes, and peak relator length. A faster certified path supports only this search policy; budget exhaustion is an inconclusive search outcome.

Bridge still needed. Even a new path between hard presentations need not trivialize either. General progress would require a structural reduction valid for all balanced trivial-group presentations, currently unprovided.

Established / proposed. Explicit move certificates and finite search frontiers already exist; learned bidirectional prioritization is a proposed extension whose originality has not been assessed.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2607.23611

059 Slice-ribbon conjecture

Simulated knot concordance researcher

Target. Every smooth knot K in S3 that bounds a smoothly embedded disk in B4 also bounds a ribbon disk in B4.

Restricted question. Does the specified slice knot 18_nh00000601 admit a ribbon-band presentation within a completely defined finite encoding and crossing budget?

Approach. Investigate the published slice knot 18_nh00000601 through bounded ribbon-band searches. Encode candidate band presentations combinatorially and verify the resulting boundary knot independently. Instead of treating its known slice disk as a ribbon disk, use that disk only to suggest geometric locations for candidate bands.

Challenge. No ribbon presentation within a chosen band or crossing budget is not evidence of non-ribbonness strong enough for a counterexample. Diagram identification also requires rigorous verification.

Next task. Specify band attachment conventions, a diagram code for 18_nh00000601, and at most three bands and twelve band-diagram crossings; validate both known controls before enumerating candidates.

Baseline. The cited paper supplies sliceness, not ribbonness. First verify the band encoding on the unknot and a published ribbon presentation of the stevedore knot 6_1. Keep the known sliceness proof and proposed ribbon certificate separately referenced: their geometric conditions and permissible disk descriptions differ.

Informative outcome. A checked band presentation with an independently established boundary-knot identification proves ribbonness for that knot. Exhausting the finite encoding proves only absence within that encoding.

Bridge still needed. Finding a ribbon disk for one candidate does not prove the universal implication; proving that a slice candidate is not ribbon would disprove it. Failure of bounded search supplies neither.

Established / proposed. The candidate's sliceness and ribbon-band methods are published; using its slice construction to prioritize bounded band searches is a proposed experiment.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2603.23717

060 Crossing-number additivity conjecture

Simulated knot diagram theorist

Target. For all tame knots K and L in S3, the minimum crossing number satisfies c(K connected-sum L) = c(K) + c(L).

Restricted question. Can a specified crossing-accounting potential survive every local labeled Reidemeister move while reproducing the restricted-move inequalities studied in the source? The potential itself is not yet specified.

Approach. Extend the restricted-Reidemeister-move approach by labeling diagram arcs according to their original summand. Track mixed crossings and a proposed crossing-debt quantity during unrestricted moves. Search for a local accounting rule that recovers crossings temporarily hidden by interactions between the two summands, initially for one restricted diagram class.

Challenge. Summand labels need not correspond to a canonical decomposition after arbitrary moves. A locally monotone quantity might fail on a single Reidemeister configuration or simply reproduce known lower bounds.

Next task. Write the potential's actual formula and list its local states before computation. Enumerate all labeled one-, two-, and three-strand Reidemeister configurations and retain any explicit violating move.

Baseline. The elementary upper bound comes from juxtaposing minimal diagrams. The source's restricted-move statement is the comparison result; unrestricted monotonicity is an additional, unproved requirement.

Informative outcome. One legal move violating the proposed accounting rule rejects that rule. Feasible local inequalities only identify a candidate potential and still require a globally coherent labeling.

Bridge still needed. The missing bridge is a well-defined bound valid for every diagram of every connected sum. Finite local tests and summand labels alone do not establish it.

Established / proposed. Restricted-move additivity is studied in the source; extending its accounting through mixed crossings is speculative and carries no novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2511.18016

061 Purely cosmetic surgery conjecture for knots in the three-sphere

Simulated computational three-manifold topologist

Target. For a nontrivial knot K in S3, surgeries along two distinct nonmeridional rational slopes never produce orientation-preservingly homeomorphic three-manifolds.

Restricted question. Which reduced slope pairs on knots 4_1, 5_2, 6_1, 6_2, and 6_3 survive a specified sequence of exact surgery obstructions?

Approach. Combine algebraic obstructions from Jones-polynomial derivatives with certified geometric comparisons of Dehn fillings. Prioritize slope pairs that survive the algebraic filters rather than sampling indiscriminately. For a bounded collection of hyperbolic knots, compare oriented invariants and attempt explicit distinguishing certificates, treating exceptional fillings separately.

Challenge. Matching computed invariants does not prove homeomorphism, while floating-point geometric differences need certification. Even a complete finite table would leave infinitely many knots and surgery slopes untreated.

Next task. List reduced p/q with $|p| \leq 4$ and $1 \leq q \leq 4$, normalize orientations, and specify each obstruction's hypotheses before analyzing the five knots. Record failed certifications separately.

Baseline. Use homology first, then applicable Jones-derived obstructions from the source. Only surviving hyperbolic fillings proceed to certified geometric comparison; exceptional fillings require separate tools.

Informative outcome. The output is a pair-by-obstruction table with certificates and explicit unknowns. Equal invariants or failed geometric certification leave a pair unresolved, rather than cosmetic. Preserve slope labels and orientation throughout the comparison.

Bridge still needed. The full target ranges over all knots and slopes. A reusable symbolic obstruction covering an infinite family would be a bridge; a finite table is calibration.

Established / proposed. Polynomial-derived surgery obstructions are established; combining them into a certificate-producing triage pipeline is the suggested extension, with no originality asserted.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2512.20511

062 Volume conjecture

Simulated quantum topologist and asymptotic analyst

Target. Hyperbolic-knot version: for every hyperbolic K in S3, lim as N→∞ of (2π/N) log|J_N(K;exp(2πi/N))| equals the hyperbolic volume of its complement; J_N(unknot)=1.

Restricted question. For the figure-eight knot, do explicitly derived finite-color correction terms improve the residual against the established asymptotic expression without numerical cancellation artifacts?

Approach. Treat finite-color data as a guide to saddle-point selection rather than fitting volume alone. For a knot with an explicit colored-Jones formula, compare competing complex saddle contributions and their cancellations. Seek an error-controlled asymptotic expansion that explains both the geometric volume and the observed finite-color corrections.

Challenge. Excellent numerical volume predictions cannot establish the required limit. Complex phase cancellations and changes of dominant saddle can invalidate a plausible asymptotic expansion unless controlled uniformly.

Next task. Locate and normalize the two figure-eight formulas, derive one explicit correction, then evaluate colors 20-80 at 80 and 160 decimal digits. Stop if the overlap digits disagree.

Baseline. The figure-eight case is a solved calibration example. Compare the derived correction with the leading limit and reproduce evaluations from two equivalent exact formulas before numerical fitting. This is calibration only.

Informative outcome. Precision-sensitive residuals or disagreement between formulas reject the numerical pipeline. Stable residual decay tests the chosen correction on one known knot, not the universal limit.

Bridge still needed. A proof for further knots requires uniform asymptotic error control, justified saddle selection, and treatment of phase cancellation. Numerical extrapolation supplies none of those automatically.

Established / proposed. Colored-Jones asymptotics and finite-color volume prediction are established approaches; using residuals to prioritize rigorously controlled saddle analysis is a proposed extension.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2502.18575

063 11/8 conjecture

Simulated equivariant stable homotopy theorist

Target. Every closed connected oriented smooth spin four-manifold M satisfies b2(M) ≥ (11/8)|signature(M)|.

Restricted question. For forms 2k(-E8) plus l hyperbolic planes with $1 \leq k \leq 8$, which $l < 3k$ remain outside the cited equivariant obstructions?

Approach. Organize potential intersection forms as 2k(-E8) plus l hyperbolic planes, where the conjecture requires l at least 3k. Compare the exact published Pin(2)-equivariant obstruction data with the remaining numerical region. Investigate whether additional geometric constraints on realizable monopole maps exclude cases that abstract equivariant maps alone permit.

Challenge. The source already completely analyzes a relevant equivariant map-existence problem. Recomputing those obstructions cannot close the remaining gap; genuinely additional geometric information would be required.

Next task. Transcribe theorem hypotheses and the obstruction table, cross-check K3 conventions, and enumerate $k \leq 8$ with $0 \leq l < 3k$. Mark every exception rather than extrapolating across it.

Baseline. Use the source's exact obstruction results and exceptions, not a simplified slogan about a 10/8+4 bound. Standard K3 and connected-sum examples check the convention. Distinguish an unexcluded intersection form from a proved smooth realization of that form.

Informative outcome. A reconciled table distinguishes excluded forms, realized forms, and cases left unexcluded. An inconsistency on known examples rejects the transcription before proposing any stronger obstruction.

Bridge still needed. Abstract equivariant-map obstructions have a demonstrated limitation. Eliminating the remaining forms requires additional geometry of realizable monopole maps, and the complete conjecture is broader still.

Established / proposed. The Pin(2) obstruction calculation is established; seeking restrictions distinguishing geometric monopole maps from abstract equivariant maps is a speculative direction, not a new result.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1812.04052

064 Borel rigidity conjecture

Simulated geometric surgery theorist

Target. Every homotopy equivalence between closed connected aspherical topological manifolds is homotopic to a homeomorphism.

Restricted question. Can a gluing ledger for two explicit aspherical pieces identify all boundary, torsion, and splitting assumptions needed to assemble a known rigidity argument?

Approach. Study closed aspherical manifolds assembled by gluing pieces along incompressible boundaries. Replace the vague goal of global rigidity with an explicit ledger of boundary identifications, Whitehead torsion, and surgery obstructions. Examine whether compatible rigidity certificates for individual pieces can be glued without losing control of the boundary maps.

Challenge. Rigidity of pieces does not automatically imply rigidity after gluing. Splitting obstructions and compatibility conditions may contain exactly the unresolved information, so the ledger alone proves nothing.

Next task. Write the T5 gluing maps and relevant fundamental-group inclusions, then attach precise source theorem references to each rigidity step. Stop at the first unchecked splitting hypothesis.

Baseline. Start from two copies of T4 × [0,1] joined cyclically by identity boundary maps, giving T5. This is a solved model, not a candidate unknown manifold. Fix dimension five throughout this calibration.

Informative outcome. A complete ledger exposes exactly where a theorem is invoked and whether its dimension and group hypotheses apply. An unmet hypothesis is an obstruction to this argument only.

Bridge still needed. Replacing the model’s simple pieces with arbitrary aspherical manifolds demands new splitting and assembly control. Compatibility of a finite example cannot establish topological rigidity in general.

Established / proposed. Surgery, splitting, and assembly methods are conventional; a certificate ledger for a fixed gluing construction is a proposed organizational experiment without novelty claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2507.11337

065 Novikov conjecture

Simulated higher-index theorist

Target. For every closed oriented manifold M, map f:M→Bπ, and rational cohomology class a on Bπ, the pairing of L(M) with f*a is invariant under orientation-preserving homotopy equivalence.

Restricted question. Can one explicit degree-two higher-signature calculation be written with all classifying-map and orientation data, then used to identify the extra requirements of an extension argument?

Approach. Fix a group with an explicit finite classifying space and study individual rational higher signatures instead of the full assembly map at once. Couple symbolic cohomology calculations with controlled geometric representatives. Ask whether a restricted collection of cohomology classes admits direct homotopy-invariance certificates that remain meaningful under a chosen group extension.

Challenge. Computable cohomology classes do not provide analytic control of the assembly map. Invariance for a few classes, especially classes inherited from known cases, may add no progress toward the general conjecture.

Next task. Specify a cell decomposition and orientation cocycle for Σ2, calculate the pairing exactly, and write an extension-dependency checklist before choosing any further group.

Baseline. For a closed genus-two surface Σ2, use its fundamental group and the identity classifying map. The degree-two pairing with its orientation class is one, a solved control.

Informative outcome. A sign or normalization discrepancy falsifies the implementation. Correct reproduction only validates the pairing conventions; a proposed extension must name a class not already handled by the benchmark.

Bridge still needed. General homotopy invariance requires appropriate topological or analytic assembly control. Computing finitely many cohomology pairings cannot establish injectivity for arbitrary groups and all relevant classes.

Established / proposed. Higher signatures and assembly-based invariance are established; class-by-class certificate extraction is a proposed preparatory workflow rather than a claimed new method.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2507.11337

066 Farrell-Jones conjecture

Simulated algebraic K-theorist

Target. For every discrete group G and unital ring R, the virtually-cyclic assembly maps computing nonconnective algebraic K-theory and ultimate lower L-theory of R[G] are isomorphisms in all degrees, with involution supplied for L-theory.

Restricted question. For a fixed graph of groups, can an auditable dependency graph decide which published inheritance theorems already settle its assembly conjecture?

Approach. Build a dependency graph for a specified group construction, recording exactly which inheritance properties reduce its assembly conjecture to simpler groups. For a graph of groups, separate vertex groups, edge groups, and virtually cyclic subgroups. Search for one missing controlled-algebra lemma rather than treating the assembly isomorphism as a monolithic target.

Challenge. The conjecture is already known for extensive group classes, and an apparent gap may merely reflect an overlooked theorem. Reduction graphs cannot manufacture an unavailable inheritance principle.

Next task. Draw the stated graph of groups, attach theorem numbers from the survey to each deduction, and record ring assumptions. Only after that audit select a genuinely unresolved construction.

Baseline. Use one vertex group Z and one trivial loop-edge group, whose fundamental group is Z * Z. Existing results settle this free-group model and validate the audit.

Informative outcome. Recovering the solved model checks the transcription of closure properties. A residual node indicates an unverified hypothesis, not necessarily an open case until a closer literature check.

Bridge still needed. The general conjecture needs arguments for groups not covered by proven inheritance mechanisms. A dependency diagram does not supply a missing controlled-algebra theorem or establish its hypotheses.

Established / proposed. Inheritance and controlled assembly arguments are established; automated dependency auditing is a suggested research aid with no claim to mathematical originality.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2507.11337

067 Inverse Galois problem over the rational numbers

Simulated arithmetic geometer

Target. Every finite group G is isomorphic to Gal(L/Q) for some finite Galois extension L of the rational numbers.

Restricted question. Can a branch-cycle and specialization workflow reproduce one rational realization while explicitly recording which arithmetic descent claims cannot be inferred from complex monodromy?

Approach. Target one finite group whose realization over the rationals is first confirmed unresolved. Enumerate feasible branch-cycle tuples, then use arithmetic constraints to prioritize candidate covers of the projective line. Treat the difficult descent to rational coefficients as the central problem, keeping complex realizability and rational realizability explicitly separate.

Challenge. A suitable tuple can establish a complex cover without supplying a rational model. Descent obstructions and specialization conditions are substantive arithmetic barriers, not details solvable by polynomial fitting.

Next task. Verify x^3-2 by exact arithmetic, identify the rational cover used for any branch-cycle comparison, and write the descent requirements before enumerating tuples for another group.

Baseline. Use x^3-2: it is irreducible over Q and has nonsquare discriminant -108, so its splitting field has Galois group S3. This is a known calibration.

Informative outcome. A certificate should distinguish irreducibility, discriminant, permutation action, and rational descent. Failure to justify descent rejects a candidate cover as a rational realization, not as a complex cover.

Bridge still needed. After calibration, a new target requires a verified unresolved group and rational model. Complex realizability plus a guessed polynomial does not bridge to every finite group over Q.

Established / proposed. Rigidity, branched covers, and specialization are classical methods; constraint-guided prioritization with an explicit descent audit is a suggested extension, not a novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1512.08708

068 Tate conjecture

Simulated arithmetic geometry computationalist

Target. Cycle-class version over finite fields: for every smooth projective X/Fq, prime ℓ not dividing q, and codimension r, algebraic cycles defined over Fq with Qℓ coefficients span the Frobenius-fixed part of H^(2r)(Xbar,Qℓ(r)).

Restricted question. Can point counts and explicit divisor intersections produce honest lower and upper rank information for one concretely specified surface, without reconstructing unsupported Frobenius eigenvalues?

Approach. For a fixed smooth projective surface over a finite field, maintain two independent estimates: Frobenius information from point counts and the rank generated by explicitly constructed divisors. Let any certified gap guide searches for curves on finite field extensions. Track which conclusions need the full characteristic polynomial rather than a few numerical traces.

Challenge. A missing explicit divisor need not imply a failure of Tate. Short point-count sequences may not determine the relevant Frobenius eigenspace, and rank comparisons require certified independence.

Next task. Check the split quadric over extension degrees one through three and its two rulings; then select a quintic equation, prove smoothness, and budget the additional point counts required.

Baseline. Begin with the smooth split quadric x0x1-x2x3=0 over F7. Its two ruling classes and known point-count formula provide an exact solved surface benchmark.

Informative outcome. Counts inconsistent with (7^e+1)^2 or an incorrect ruling intersection matrix reject the pipeline. Three traces on a later quintic alone cannot certify its entire invariant eigenspace.

Bridge still needed. Progress beyond calibration needs enough certified Frobenius data and algebraic cycles to close a justified rank gap. Surfaces and divisors address only part of the stated cycle-class target.

Established / proposed. Point counts, Frobenius, and divisor intersection calculations are standard; using their certified mismatch to prioritize curve searches is an experimental proposal without novelty claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2505.07337

069 Lefschetz standard conjecture

Simulated algebraic cycles researcher

Target. Complex version of the Lefschetz standard conjecture: for every smooth complex projective d-fold X with ample class h and i ≤ d, the inverse of h^(d-i):H^i(X,Q)→H^(2d-i)(X,Q) is induced by an algebraic correspondence. Arbitrary-characteristic formulations are outside this card.

Restricted question. Can the inverse maps on an explicit product of elliptic curves be represented by named algebraic cycles, rather than only by abstract rational matrices?

Approach. Work on a fixed smooth projective variety with an explicit cohomology basis and known correspondences. Express the inverse hard-Lefschetz map as a rational matrix, then compare it with the span of actions of algebraic correspondences already available. Study which additional family of cycles could supply any missing matrix directions.

Challenge. Linear algebra can reveal that a chosen cycle library is insufficient, but cannot show that suitable algebraic cycles do not exist. Producing missing correspondences is precisely the difficult part.

Next task. Fix symplectic bases of H1(E,Q), compute the polarized cup-product matrices, and identify product correspondences realizing their inverses; specify cycle-action conventions before extending the library.

Baseline. Take E:y^2=x^3-x with its smooth projective completion, X=E×E, and the product polarization. The case is solved; its product cycles calibrate correspondence conventions.

Informative outcome. Matrix equality without an actual cycle is insufficient. Useful calibration lists each correspondence, its degree, and its induced action; any mismatch identifies a normalization error.

Bridge still needed. On an unknown variety, a missing matrix direction requires constructing an algebraic cycle, the central difficulty. A finite cycle library cannot establish that no such cycle exists.

Established / proposed. Correspondence actions and hard-Lefschetz linear algebra are established; organizing candidate cycles by missing matrix directions is a suggested experiment, not a new theorem.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1002.5011

070 Abundance conjecture

Simulated birational geometer

Target. For every projective klt pair (X,Δ) in characteristic zero with rational boundary, if the Q-Cartier divisor KX+Δ is nef, some positive integral multiple is base-point free.

Restricted question. For smooth projective fourfolds with nef canonical divisor of numerical dimension two, which exact nonvanishing input would support the proposed fibration reduction?

Approach. Restrict to projective klt fourfold pairs with nef log-canonical divisor and a specified numerical dimension. Translate semiampleness into the search for sections with controlled base loci. Compare numerical data with candidate fibrations, then isolate exactly which nonvanishing statement would permit a canonical-bundle-formula reduction to lower dimension.

Challenge. Numerical positivity does not itself produce sections. Assuming the required fibration or nonvanishing would already import a central unresolved ingredient, so every reduction must expose rather than conceal that assumption.

Next task. Calculate the benchmark's section growth and canonical map, then list every product-dependent step. Stop the generalization where a section or fibration is assumed rather than constructed.

Baseline. Use X=C×C×E×E, with smooth projective C:y^2=x^5-1 and E:y^2=x^3-x. Projection to C×C explains its known semiample canonical divisor. Both curves mean their smooth projective completions over C.

Informative outcome. The benchmark should recover numerical dimension two and, for m≥2, section dimension (2m-1)^2. A reduction needing an assumed section is recorded as conditional rather than proved.

Bridge still needed. A non-product fourfold needs an independently justified nonvanishing or fibration theorem. The full target additionally includes all dimensions and klt boundaries, neither supplied by product calculations.

Established / proposed. Nonvanishing, canonical bundle formulas, and semiampleness are conventional tools; tracing their exact dependency in a restricted fourfold class is a proposed exploratory step.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2406.18233

071 Termination of flips conjecture

Simulated minimal-model-program specialist

Target. For any projective klt pair (X,Δ) in characteristic zero with rational boundary, every sequence of $(KX+\Delta)$-flips terminates.

Restricted question. Can an explicitly defined lexicographic quantity decrease along a fixed collection of genuine flips and take values in a proven well-founded set?

Approach. Search for a lexicographic progress measure combining discrepancy data with changes in the negative part of the Nakayama-Zariski decomposition. Restrict to an explicitly specified class of projective klt pairs. Use known flip sequences to falsify candidate monotonicity rules before attempting a proof that any surviving measure is well founded.

Challenge. Strict decrease of a real-valued quantity does not guarantee termination. The proposed measure needs both monotonicity under every allowed flip and a genuinely well-founded range, which are substantial missing proofs.

Next task. Specify three fan-level examples and the actual candidate formula before evaluation; verify each is a flip, not a flop, and include a proof obligation for the value set’s well-foundedness.

Baseline. Use verified toric examples with fans, boundary coefficients, and flip directions specified in advance. Their known termination calibrates bookkeeping; it is not evidence of a universal measure.

Informative outcome. An increase rejects monotonicity. A decreasing real sequence with no well-foundedness proof also fails as a termination argument, even if all chosen finite sequences end.

Bridge still needed. The bridge requires both universal monotonicity for the selected class and a well-founded range, followed by extension beyond that class. Neither follows from sampling explicit flip sequences.

Established / proposed. Discrepancy and Nakayama-Zariski approaches are established; experimentally combining them into a well-founded progress measure is speculative and not claimed original.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2305.01752

072 Nagata’s conjecture on plane curves

Simulated interpolation and algebraic geometry researcher

Target. For $n \geq 10$ very general points in $P2(C)$, every nonzero plane curve of degree d with multiplicities m_i satisfies $d\sqrt{n} > \Sigma m_i$.

Restricted question. Can exact rank certificates verify the equal-multiplicity, ten-point systems with $1 \leq m \leq 6$ and $d = \text{floor}(\sqrt{10}\, m)$, using explicit rational configurations?

Approach. Focus on ten very general points with equal multiplicity, the smallest nonsquare count in the conjecture's range $n >= 10$. Build exact interpolation matrices at carefully chosen rational configurations and certify full rank below the predicted degree threshold. Compare certificates across multiplicities to seek a degeneration pattern that could replace separate computations with an inductive argument.

Challenge. Each full-rank certificate proves only a particular finite interpolation case by openness. Arbitrarily many tested multiplicities still do not establish the conjecture's universal degree and multiplicity inequality.

Next task. Test points $(i, (i^2 + i^5) \bmod 101)$, $i = 0, \ldots, 9$, interpreting coordinates as ordinary integers. For $m = 1, \ldots, 6$ set $d = \text{isqrt}(10m^2)$, assemble exact matrices, and cap elimination at thirty minutes.

Baseline. Use vanishing of every partial derivative of total order below m as the interpolation conditions. Full column rank at one rational configuration implies the corresponding very-general system is empty.

Informative outcome. Save rational pivot certificates and rank-nullity checks. Rank deficiency at the chosen special configuration is inconclusive; a second configuration tests whether it was caused by the sample.

Bridge still needed. Finite multiplicities do not imply an asymptotic inequality. A genuine bridge would be a justified degeneration or induction controlling all multiplicities and eventually all nonsquare point counts.

Established / proposed. Interpolation and degeneration arguments are established; comparing exact rank certificates to suggest an induction is an exploratory extension with no novelty claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/html/1202.0475v1

073 Segre-Harbourne-Gimigliano-Hirschowitz conjecture

Simulated specialist in linear systems of plane curves

Target. For every nonempty plane system $D=dH-\Sigma m_iE_i$ on the blowup X of $P2(C)$ at very general points, $H1(X,O(D))>0$ exactly when a smooth rational curve E satisfies $E^2=-1$ and $D\cdot E\leq -2$.

Restricted question. For ten-point systems with degree ≤ 6 and multiplicities one or two, can exact ranks and exceptional fixed components explain every observed departure from expected dimension?

Approach. Create a bounded atlas of plane interpolation systems with unequal multiplicities at very general points. Compute expected dimensions, subtract explicitly detected exceptional fixed components, and compare the adjusted prediction against certified matrix ranks. Concentrate on the first configurations where naive dimension counting differs from the corrected geometric prediction.

Challenge. A special rational point configuration can create extra curves absent for very general points. Failure to find full rank is inconclusive unless the symbolic generic rank is established.

Next task. Fix rational configurations and list the multiplicity vectors before computing; specify exceptional-curve detection and the proof of each residual dimension. Reclassify unsupported rank deficiencies as unknown.

Baseline. Expected projective dimension is $\max(-1,d(d+3)/2-\Sigma m_i(m_i+1)/2)$. For nonempty systems, speciality means nonzero $H1$; the displayed exceptional curve forces a repeated fixed component. Empty systems are nonspecial by convention.

Informative outcome. A nonzero minor certifies a generic lower rank bound; a special rational configuration does not certify generic rank deficiency. Record separately which dimension equalities have both bounds justified.

Bridge still needed. The finite atlas needs a uniform classification of all special systems before reaching SHGH. Replacing that classification by numerical nefness would silently weaken the target.

Established / proposed. Expected dimensions, exceptional curves, and interpolation matrices are classical; a certificate-backed discrepancy atlas is a proposed research aid, not an assertion of originality.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/html/1202.0475v1

074 Hopf’s positive-curvature Euler-characteristic conjecture

Simulated Riemannian topologist

Target. Every closed even-dimensional Riemannian manifold with strictly positive sectional curvature has strictly positive Euler characteristic.

Restricted question. In dimension six, can algebraic curvature tensors disprove a proposed pointwise inequality before an integrated Euler-form argument is attempted?

Approach. Test whether curvature positivity can support an integrated Euler-form inequality without demanding pointwise positivity of its density. In dimension six, use algebraic curvature tensors to stress-test candidate inequalities involving curvature invariants and divergence terms. Preserve the distinction between realizable global metrics and tensors satisfying only local algebraic constraints.

Challenge. Positive sectional curvature need not force the Euler density positive at each point. Algebraic tensor experiments cannot establish the global differential identities or closed-manifold estimates needed for integration.

Next task. Specify one tensor family and an actual proposed Euler-form inequality, then define exact positivity certification on decomposable two-vectors. Do not replace this with positivity on a sampled plane set.

Baseline. Constant-curvature tensors provide normalization checks for the Euler density. They do not test variable-curvature derivative terms or establish the global manifold identities required by an integrated estimate.

Informative outcome. A certified positive-sectional tensor violating a candidate inequality rejects that inequality. Passing a finite tensor family only leaves it unrefuted; realizability and integration remain separate obligations. Report necessary-condition failures independently from any global geometric conclusion.

Bridge still needed. The conjecture concerns global closed metrics. An integrated inequality with justified divergence terms and topology is needed; local algebraic inequalities alone cannot establish its conclusion.

Established / proposed. Curvature identities and Euler-characteristic methods are standard; tensor-based falsification of candidate integrated inequalities is a proposed preparatory experiment without novelty claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2507.16936

075 Hopf’s S2 × S2 problem

Simulated differential geometer

Target. There is no smooth Riemannian metric on S2×S2 having strictly positive sectional curvature on every tangent two-plane.

Restricted question. For an explicitly bounded family of conformal deformations of the round product metric, do the mixed-plane positivity constraints admit any small parameter choice? This question concerns an ansatz, not every nearby metric.

Approach. Use a restricted family of smooth deformations of the round product metric as a controlled obstruction laboratory. Track sectional curvature specifically on formerly flat mixed planes, including second-order terms. Search for an integral compatibility constraint forcing some mixed plane to remain nonpositive within the chosen deformation family.

Challenge. Failure of deformations near the product metric would not exclude distant positively curved metrics. First-order curvature tests can miss second-order effects, and sampled planes cannot certify positivity everywhere.

Next task. Set $g_\varepsilon = \exp(2\varepsilon f) g_0$ with f a specified linear combination of spherical harmonics of degrees ≤ 2 on each factor; derive curvature and remainder bounds before checking parameter feasibility.

Baseline. The product metric has zero sectional curvature on mixed planes. Known nearby-obstruction ideas are context; a finite conformal ansatz is a deliberately restricted diagnostic.

Informative outcome. An exact incompatible inequality system rejects that ansatz. A numerically feasible point still needs uniform curvature certification; truncated second-order positivity cannot certify the actual metric.

Bridge still needed. Failure near the product says nothing about distant or nonconformal metrics. Reaching the full target needs a global obstruction independent of this restricted deformation family.

Established / proposed. Product-metric deformation and curvature obstruction approaches are established; a certified low-mode feasibility study is a bounded proposed experiment, not a global solution or novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/math/0411632

076 Chern’s isoparametric conjecture for minimal hypersurfaces

Simulated minimal-submanifold analyst

Target. Strong Chern formulation: for every n≥2, each connected closed minimal immersed hypersurface M^n in the unit sphere S^(n+1) with constant scalar curvature has constant principal curvatures, hence is isoparametric.

Restricted question. For M4 in S5, can the derivative terms of K=det(A) be controlled after dropping the additional assumption that K is constant?

Approach. Restore the K-derivative terms in one identity from the four-dimensional rigidity argument instead of imposing constancy of K. Build a symbolic necessary-condition model of curvature jets and ask whether the proposed estimate survives nonzero ∇K. Keep isoparametric examples only as zero-derivative regression controls.

Challenge. The restored terms may have indefinite sign. A locally permitted jet need not extend to a global closed hypersurface, and vanishing-derivative examples cannot test the obstacle under investigation.

Next task. Extract one source identity before setting ∇K=0; introduce symbolic A and covariant derivatives, impose Gauss-Codazzi and constant trace/norm identities, and test nonzero K-derivative terms explicitly.

Baseline. The cited rigidity result assumes constant K. Isoparametric examples have ∇K=0, so they only calibrate the old identity and cannot test the additional terms.

Informative outcome. A permitted symbolic jet making the proposed bound fail rejects that bound. Successful jet filtering neither constructs a global hypersurface nor proves the needed integrated estimate.

Bridge still needed. The required bridge is an inequality justified by the original hypotheses, including all derivative terms and global integration. A non-isoparametric global example must not be assumed as input.

Established / proposed. The extra constant-K result is established input; controlling nonzero derivative terms is a proposed extension. The constant-principal-curvature definition is documented in arXiv:2411.04231; no novelty is asserted.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2606.29246

https://arxiv.org/abs/2411.04231

077 Symmetric Mahler conjecture in dimensions four and higher

Simulated convex geometer

Target. For every origin-symmetric convex body K in Rn with nonempty interior and n≥4, volume(K)·volume(K polar) ≥ 4^n/n!.

Restricted question. For specified centrally symmetric ten-vertex rational four-polytopes, can a one-parameter vertex motion preserve feasibility while lowering the exact volume product?

Approach. Concentrate on centrally symmetric rational polytopes in four dimensions that lack coordinate reflection symmetry. Compute their volume products exactly and study symmetry-preserving vertex motions toward simpler polytopes. Seek a rigorously nonincreasing deformation rule within one fixed combinatorial class, allowing several equality shapes rather than assuming the cube is unique.

Challenge. Optimizing a bounded polytope class does not control all convex bodies. Polar combinatorics can change during deformation, and equality includes more shapes than just cubes, complicating a universal descent rule.

Next task. Define twenty exact ten-vertex inputs and test v5→v5+t e1 with t=-1/4,0,1/4, maintaining antipodes. Specify triangulation, polar construction, and degeneracy handling before any search.

Baseline. In dimension four the bound is 32/3, attained by the cube and cross-polytope among other examples. Both provide normalization checks before non-unconditional shapes are tested.

Informative outcome. A certified product below 32/3 would demand immediate independent verification. A decreasing path remaining above it suggests only an ansatz-level descent; combinatorial transitions must be logged.

Bridge still needed. A full argument needs a valid reduction covering all symmetric convex bodies, not just ten-vertex polytopes. A restricted descent rule is not automatically such a reduction.

Established / proposed. Volume products and deformation methods are established; exact deformation testing on a specified non-unconditional polytope class is an exploratory extension without novelty claims.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://video.renyi.hu/video/artem-zvavitch-volume-product-and-mahler-conjecture-for-convex-bodies-407

078 Zariski cancellation problem in characteristic zero

Simulated affine algebraic geometer

Target. Complex affine-space version: for every complex affine variety X, an isomorphism X×A1 ≅ A^(n+1) implies X ≅ A^n; the proposed focus is n=3.

Restricted question. Which derivation data can distinguish an affine-space cylinder from cylinders that are merely isomorphic to one another over nonisomorphic base varieties?

Approach. Fix complex affine threefolds and analyze what a proposed cylinder isomorphism would do to derivations and invariant subrings. Use examples where cancellation genuinely fails for more general varieties as adversarial controls. Search for an additional property distinguishing cylinders that are affine four-space from cylinders merely isomorphic to each other.

Challenge. Counterexamples to cancellation between general varieties do not answer affine-space cancellation. Known derivation invariants can collapse after adjoining a variable, so transferring them back is exactly the difficult step.

Next task. Identify one survey example by equations and proposition, write its cylinder isomorphism, and compute the specified locally nilpotent derivation data. Stop if the required maps are unavailable.

Baseline. The survey distinguishes positive-characteristic counterexamples and characteristic-zero questions. A published characteristic-zero pair with isomorphic cylinders is a control, not a counterexample to the selected affine-space statement.

Informative outcome. A useful audit identifies the exact cylinder isomorphism and where a chosen invariant changes under stabilization. Invariant collapse rejects that invariant as the proposed automatic descent mechanism.

Bridge still needed. The missing step is recovering affine-space structure from the cylinder assumption. General cancellation failures and unsuccessful invariant calculations do not furnish a counterexample to that implication.

Established / proposed. Locally nilpotent derivations and cancellation examples are standard tools; adversarially separating general cancellation from affine-space cancellation is a proposed diagnostic experiment.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2208.14736

079 Bounded negativity conjecture in characteristic zero

Simulated algebraic surface geometer

Target. For every smooth projective complex surface X, there exists b(X) such that $C^2 \geq -b(X)$ for every reduced irreducible curve C on X.

Restricted question. On one fixed ten-point blowup, which bounded numerical curve classes survive adjunction and exact interpolation, and which surviving sections can actually be proved irreducible?

Approach. Fix one smooth projective surface obtained by blowing up a specified configuration of complex plane points. Enumerate candidate curve classes with increasingly negative self-intersection, then filter them using adjunction, intersection constraints, and exact interpolation. Study which obstruction repeatedly prevents numerical classes from being represented by irreducible curves.

Challenge. Numerical classes with very negative square need not contain any curves. Bounds observed across different blowups do not establish a bound on one fixed surface, and reducibility must be addressed.

Next task. Publish ten rational coordinates, enumerate $d \leq 8$ and $0 \leq m_i \leq d$ with a stated cap, and factor selected sections exactly. Do not equate a nonzero interpolation kernel with irreducibility.

Baseline. Keep the ten point coordinates fixed for the entire experiment. Compare numerical square and genus constraints with actual linear-system sections; neither alone establishes existence of an irreducible curve.

Informative outcome. The output separates impossible classes, effective reducible examples, certified irreducible examples, and unknowns. A sequence on changing blowups cannot demonstrate unbounded negativity on one surface.

Bridge still needed. Even a very negative irreducible curve only improves a lower benchmark for b(X). The full conjecture needs a bound for every curve, independent of degree and multiplicity.

Established / proposed. Blowups, adjunction, and interpolation are established methods; tracking the precise obstruction for bounded candidate classes is a proposed experiment, not a claimed new bound.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2101.07187

080 Four-dimensional sphere-packing problem

Simulated discrete geometer and optimization researcher

Target. Determine the maximal upper density of equal-ball packings in R4; the D4-optimality target is density $\leq \pi^2/16$ for every packing, with equality attained by D4.

Restricted question. Can a certified periodic-packing pipeline reproduce D4's known local obstruction while supporting a separately specified finite-amplitude search over lattice and translation variables?

Approach. Separate validation from discovery. Local multi-translate perturbations of D4 should reproduce periodic extremality. A discovery search would instead vary both a lattice basis and translation vectors at finite amplitude from several starts. Independently, reproduce an unrestricted three-point upper bound, keeping lattice-only inequalities explicitly separate.

Challenge. Periodic extremality already rules out a sufficiently local improvement. Finite-amplitude searches need certified neighbor constraints and global cell control; their failure does not prove unrestricted optimality.

Next task. Specify P=BZ4+{0,t}, a compact nonsingular box for B and t, and a certified finite neighbor cutoff. Separate small D4 calibration from multistart finite-amplitude candidates before optimization. Mark all still-undefined parameter bounds as design obligations.

Baseline. Schürmann proves D4 is periodic extreme, including multiple-translate descriptions (arXiv:0808.2013, Corollary 11). Local perturbations are a no-improvement control, not the proposed discovery mechanism.

Informative outcome. A local numerical gain triggers a certification or neighborhood audit. A verified finite-amplitude denser packing would refute D4 optimality; no gain leaves the unrestricted maximum unresolved.

Bridge still needed. Finite periodic searches cannot prove the universal upper bound. Three-point inequalities must cover unrestricted packings, while local extremality supplies no global-optimality theorem.

Established / proposed. Three-point bounds and D4 periodic extremality are established. The proposed combination is untested; finite-amplitude search is distinct from the solved local question and carries no originality claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2206.15373

https://arxiv.org/abs/0808.2013

81-100 / Computation, dynamics and analysis

081 P versus BPP

Simulated combinatorics and complexity specialist

Target. Determine whether every language decided in polynomial time with independent random bits and error at most 1/3 on every input has a deterministic polynomial-time decider: P=BPP.

Restricted question. For fixed read order x1,x2,x3, two states per layer, start state 0, and arbitrary accepting subsets, minimize a uniform sampler's support subject to acceptance error at most 1/4.

Approach. Replace the vacuous existence test with exact support optimization, then test composition rather than assuming it. The completed calibration identifies a four-string optimum. A second completed test shows that independently concatenating this sampler can exceed the original error guarantee, giving a concrete obstruction for the next design.

Challenge. The obstruction concerns this fixed sampler and independent concatenation. It rules out neither all correlated samplers nor derandomization of longer branching programs, much less general randomized polynomial-time computation.

Next task. Design a correlated six-bit sampler or an explicit error-growth recurrence. Specify the joint distribution and boundary-state guarantee before execution; do not rerun the completed support enumeration.

Baseline. Uniform sampling of all eight strings is exact. The 16,384 permitted program descriptions induce 88 distinct Boolean functions; this is the complete calibration class.

Informative outcome. Completed composition failure: one permitted six-layer program accepts 40/64 full inputs but 4/16 inputs from S×S, an error of 3/8. Independent concatenation therefore fails the proposed 1/4 guarantee.

Bridge still needed. Unknown: construct uniform, efficiently computable samplers for growing program sizes with controlled accumulated error, then handle general polynomial-time computation. A finite optimum supplies no such construction.

Established / proposed. Wigderson's pseudorandomness framework is established. The finite calibrations are verified project results, with no originality or general derandomization claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result. New finite verification: minimum uniform support 4, with 24 optimizers, for the exact three-bit model at error 1/4. The fixed product sampler S×S has exact worst six-bit error 3/8; its desired fixed-error composition claim is rejected. Both results have been crosschecked.

https://www.math.ias.edu/files/Book-online-Aug0619.pdf

082 NP versus coNP

Simulated formalization specialist

Target. Determine whether NP=coNP; equivalently, whether all unsatisfiable Boolean formulas admit certificates of polynomially bounded length accepted by one polynomial-time verifier.

Restricted question. On fixed odd-charge Tseitin formulas, how much does sharing identical separator subarguments reduce a fully checkable unsatisfiability certificate relative to an unshared tree?

Approach. Make the certificate language explicit before optimizing it. A node records a separator assignment, the residual clauses, and either a recursively justified contradiction or a pointer to an identical earlier node. Compare tree and directed-acyclic representations while checking every pointer and local inference independently.

Challenge. An efficient verifier does not make its certificates short. A restricted certificate language may have exponential lower bounds even if another proof system behaves differently.

Next task. First specify canonical clause encodings, the separator-selection rule, and permitted leaf inferences. Then generate the three formulas and compare shared versus unshared certificate bytes and verifier checks.

Baseline. Use the cubic graphs K4, K3,3, and the cube. Place charge 1 at the least labeled vertex and 0 elsewhere; parity summation supplies an elementary contradiction control.

Informative outcome. A verified reduction in certificate size is informative only against the identical unshared derivation. Reject any scheme that needs an unrecorded semantic inference to validate a node.

Bridge still needed. Unknown: obtain polynomial-size certificates for every unsatisfiable input, not merely these parity formulas, using a fixed polynomial-time verifier. Resolution lower bounds alone do not separate NP from coNP.

Established / proposed. Aaronson's proof-complexity discussion supplies the established context. Tseitin contradictions and shared derivations are standard; the proposed comparison makes no novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

083 L versus NL

Simulated combinatorics and complexity specialist

Target. Determine whether every directed s-to-t reachability instance on n vertices can be decided deterministically using O(log n) working bits and read-only input, equivalently L=NL.

Restricted question. Can a stated frontier-reconstruction algorithm traverse chains of two-vertex directed layers without storing a visited set, including all recursion and counter storage?

Approach. Narrow the proposed graph family before searching all small digraphs. A frontier summary must have an explicit encoding and a reconstruction procedure, rather than being an informal set. Prove correctness for one extension by a two-vertex layer and write the exact memory recurrence before implementing a traversal.

Challenge. Recomputation may conceal a large call stack or a nonterminating traversal. Constant-width layered graphs also discard much of the difficulty of arbitrary directed reachability.

Next task. Specify an iterative algorithm and bit-level storage ledger; then test all four-vertex loopless digraphs against exhaustive reachability, treating inputs outside the layered class as adversarial checks.

Baseline. Breadth-first search gives the correctness oracle and an explicit larger-memory comparison. A simple layered dynamic program is the known restricted baseline, not a discovery.

Informative outcome. A single reachability mismatch or hidden unbounded stack rejects the implementation. Success counts as a verified restricted algorithm only after its all-layer inductive proof is supplied.

Bridge still needed. Unknown: replace the constant-width promise by an unrestricted directed graph while retaining logarithmic space. General reachability cannot be recovered by silently storing every layer or strongly connected component.

Established / proposed. Directed reachability and space accounting are established, as in Wigderson's space-complexity chapter. The reconstruction idea is a proposed audit of a restricted algorithm.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.math.ias.edu/files/Book-online-Aug0619.pdf

084 L versus BPL

Simulated probability and dynamics specialist

Target. Determine whether every polynomial-time randomized computation using O(log n) working space and error at most 1/3 has a deterministic O(log n)-space simulation: BPL=L.

Restricted question. For explicitly contracting four-state stochastic matrices, how does deterministic rounding of state probabilities affect acceptance error over sixteen steps?

Approach. Separate numerical stability from space efficiency. Study row distributions v(t+1)=v(t)P, recording an explicit contraction coefficient and the per-step rounding error. Derive an error recurrence first; use its failure near reducibility to identify which assumptions a later space-bounded simulation would actually need.

Challenge. Even a stable matrix calculation may use too much memory. General randomized computations need not contract, so this calibration cannot justify applying the recurrence universally.

Next task. For precisions b=4,8,12, floor the first three updated coordinates to multiples of 2^(-b), assign the remaining mass to coordinate four, and compare sixteen-step errors exactly. Preserve rational outputs.

Baseline. Use Pε=(1-ε)I+εJ/4 for ε in {1,1/2,1/4,1/16,0}, v0=(1,0,0,0), and accepting state 0. Exact rational powers provide ground truth.

Informative outcome. Check whether the derived bound covers every listed chain and precision. Failure rejects the bound; ε=0 is the noncontracting control, not evidence that general derandomization fails.

Bridge still needed. Unknown: replace contraction by a property valid for arbitrary configuration graphs and implement all required arithmetic within logarithmic space, with polynomial running time and uniformly controlled error.

Established / proposed. Markov representations and space-bounded derandomization are standard. This explicit chain family and rounding audit are calibration choices, not a new algorithm.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.math.ias.edu/files/Book-online-Aug0619.pdf

085 P versus PSPACE

Simulated alternative-methods researcher

Target. Determine whether P=PSPACE; a canonical equivalent question is whether every fully quantified Boolean formula can be decided in deterministic polynomial time.

Restricted question. How does quantifier-respecting elimination change reduced ordered binary-decision-diagram size on six-variable formulas, relative to their full truth tables?

Approach. Replace unspecified boundary signatures with a recognized, fully checkable representation for the baseline. Build ordered decision diagrams, eliminate the innermost variable using OR for existential quantification and AND for universal quantification, then merge identical subgraphs. Track intermediate size as well as final truth value.

Challenge. Small decision diagrams on selected formulas do not bound worst-case intermediate size. A representation may grow exponentially, even though each individual elimination step is correctly implemented.

Next task. Generate 100 formulas per prefix, each with eight distinct non-tautological clauses using seed 85; preserve variable order x1,...,x6 and compare every reduced diagram to its truth table.

Baseline. Use two prefixes, ∃x1∀x2∃x3∀x4∃x5∀x6 and its quantifier reversal, with clauses of three distinct variables. Truth-table quantification independently determines every intermediate Boolean function.

Informative outcome. Report node counts before and after each elimination, alongside exact agreement checks. A mismatch rejects the implementation; compression motivates only a clearly stated structural restriction.

Bridge still needed. Unknown: prove a polynomial representation-size and update-time bound for all quantified formulas, or invent a different representation that escapes the observed blowups without restricting alternation or input structure.

Established / proposed. Quantified formulas and decision-diagram elimination are established. The notebook reproduces a baseline before seeking any representational addition; originality is unassessed.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

086 Existence of one-way functions

Simulated number theorist

Target. Do uniformly polynomial-time computable functions f_n:{0,1}^n→{0,1}^poly(n) exist for which every probabilistic polynomial-time inverter finds any preimage of f_n(U_n) with only negligible success probability?

Restricted question. In additive Z/7Z, compare two-query deterministic inversion strategies using hidden handles for 0,1,x against the same model augmented by a challenge-parity query, with x uniform.

Approach. Choose a handle-only, Maurer-style toy baseline, not a model exposing random encoding strings. Handles reveal only insertion indices. A leakage query returns the parity of the least nonnegative integer representing x; this is deliberately algebraically correlated extra information, not a bit of an already visible label.

Challenge. This artificial leakage oracle is not a representation of ordinary access to a real cryptographic group. Any finite advantage or restricted lower bound remains far from proving one-way functions exist.

Next task. Enumerate adaptive strategies of depth at most two using Add, Subtract, and Equality on existing handles, optionally one Parity(h2) query; average success over all seven secrets, with outputs in {0,…,6}.

Baseline. Initial handles are h0=0,h1=1,h2=x. Addition or subtraction creates a fresh handle even for an existing value; equality returns one bit. Each oracle operation costs one query.

Informative outcome. Return exact optimum success fractions and attaining strategy trees for both models. Treat a copied handle-index bit as redundant leakage and require its advantage to be zero.

Bridge still needed. Unknown: obtain an asymptotic lower bound against unrestricted algorithms for an explicitly encoded function. No averaging over sampled encodings is involved here; the specified model uses abstract slot indices.

Established / proposed. Wigderson covers one-way functions; Hhan-Yamakawa-Yun §1.2 distinguishes generic models: https://arxiv.org/abs/2307.03065 . This leakage game is a diagnostic, not a security theorem.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.math.ias.edu/files/Book-online-Aug0619.pdf

https://arxiv.org/abs/2307.03065

087 Deterministic polynomial identity testing

Simulated algebra and geometry specialist

Target. Given an arithmetic circuit over the rationals, determine deterministically in time polynomial in its bit encoding whether its formal multivariate polynomial is identically zero.

Restricted question. For four-variable sums of products sharing one explicit common factor, can a prescribed Kronecker substitution verify zero identities without introducing collisions?

Approach. Use a provably valid, possibly expensive substitution as the baseline before claiming a small hitting set. Restrict total degree to six and substitute xi=t^(7^(i-1)). Record univariate degree and arithmetic cost, then separately ask whether common-factor structure permits reducing these costs uniformly.

Challenge. A factor shared by only some summands cannot be canceled from the whole sum. Kronecker substitution is collision-free under the degree bound but can become exponentially costly as variable count grows.

Next task. With seed 87 generate 100 circuits, add explicit pairs Ci-Ci, expand over Q, apply the fixed substitution, and compare coefficientwise zero tests, degrees, and operation counts.

Baseline. Let F=x1+1 and Ci=F∏j Lij, with five linear factors Lij having coefficients in {-1,0,1}; compare sums of three such terms and exact cancellation controls.

Informative outcome. Any nonzero-to-zero collision falsifies the implementation or degree accounting. Lower degrees on shared-factor examples motivate a new construction only if a uniform proof and cost bound follow.

Bridge still needed. Unknown: obtain polynomial-size, efficiently constructible tests for unrestricted circuits without exponential dependence on variables, degree, or hidden coefficient size. Successful expansion on small circuits is a calibration, not derandomization.

Established / proposed. Polynomial identity testing, common-factor manipulation, and Kronecker substitution are established. This control deliberately reproduces known behavior before a proposed efficiency improvement.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

088 VP versus VNP

Simulated algebra and geometry specialist

Target. Determine whether the permanent polynomials over Q require arithmetic circuits of superpolynomial size in matrix dimension, equivalently VP≠VNP over Q.

Restricted question. How large can a coefficient-flattening rank become for simple products of sums, relative to their arithmetic-circuit gate count?

Approach. Turn the broad search for a complexity measure into a falsification exercise. Fix the variable split X={x1,x2}, Y={y1,y2}; use the rank of the coefficient matrix indexed by X- and Y-monomials. Compare its growth with actual sharing-aware circuit size before suggesting any lower-bound inequality.

Challenge. A high flattening rank is compatible with a small circuit, particularly for products with reused subexpressions. Restricted-circuit lower bounds would not separate unrestricted VP and VNP.

Next task. Construct those matrices over Q, compute exact ranks and stated gate counts, and check candidate claims rank≤size and log 2(rank)≤size. Report counterexamples rather than fitting a favorable bound.

Baseline. Study fk=(x1+y1)^k(x2+y2)^k for k=1,...,6. Exact symbolic expansion supplies flattening matrices; exponentiation by squaring supplies an explicit circuit-size upper bound.

Informative outcome. A violated inequality is a useful rejection of that specific measure-to-size claim. An inequality surviving six cases must still pass operation-by-operation proof obligations and cancellation controls.

Bridge still needed. Unknown: find a measure small for every polynomial-size general circuit yet superpolynomially large on the permanent, with proofs covering sharing, arbitrary constants, and cancellation.

Established / proposed. Coefficient flattenings and arithmetic circuit measures are established. The product family is an adversarial control, not a proposed hard polynomial or a novelty claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

089 Efficient classical integer factoring

Simulated number theorist

Target. Deterministic formulation: can every positive integer N be factored classically in time polynomial in its binary length? Separately, does such a randomized classical algorithm exist with success probability at least 2/3 for every N?

Restricted question. How does a specified partial-information factoring algorithm trade known high bits of a factor against remaining candidate-prefix enumeration and lattice costs?

Approach. Treat the unknown information as part of the input cost. Fix a small-root theorem and an implementation before generating favorable examples; record every hypothesis, lattice parameter, and failed recovery. Compare the advertised recovery cost with the cost of supplying the assumed factor prefix when no bits are known.

Challenge. A partial-information theorem does not obtain its own missing bits. Tiny semiprimes and conveniently selected lattice parameters can hide exponential enumeration or an asymptotically unsuitable implementation.

Next task. Select and cite the concrete small-root theorem, fix lattice parameters and prime-generation rules, and publish the list of revealed-prefix lengths before running any of the proposed fifty cases.

Baseline. Trial division gives a correctness oracle at the proposed small sizes. Separately reproduce a published partial-information theorem at parameters satisfying its exact assumptions.

Informative outcome. The first deliverable is a complete theorem-to-code specification. Later results must include failed cases and total prefix-search cost; recovery after free information is not counted as an unconditional improvement.

Bridge still needed. Unknown: eliminate the external factor information with a polynomial-time procedure and prove a polynomial worst-case bound for all integers, including unbalanced or repeated factors.

Established / proposed. Aaronson discusses classical factoring complexity. Lattice partial-information recovery is established, but its exact source and implementation must be selected before any candidate addition is evaluated.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

090 Polynomial-time graph isomorphism

Simulated combinatorics and complexity specialist

Target. Determine whether isomorphism of any two finite simple undirected graphs can be decided deterministically in time polynomial in their adjacency-matrix input size.

Restricted question. Does counting edges among common neighbors distinguish the Shrikhande graph from the four-by-four rook graph after ordinary color refinement fails?

Approach. Specify the invariant rather than promising a new signature family. For each edge uv, count edges in the induced graph on N(u)∩N(v), then compare the multiset of these counts. This inexpensive control tests exact invariant implementation and relabeling behavior before any canonical-branching research.

Challenge. Distinguishing this known pair is predetermined and not a novel graph-isomorphism result. Other graph families can defeat the invariant, and no polynomial bound on canonical branching follows.

Next task. Construct both adjacency matrices; compute edge-signature multisets exactly and repeat under 100 seeded vertex permutations. Include the unchanged graph and an isomorphic relabeling as acceptance controls.

Baseline. Both graphs have vertices $Z4^2$. Rook adjacency means one coordinate agrees; Shrikhande adjacency uses differences ±(1,0), ±(0,1), ±(1,1). Uniform-start color refinement retains one vertex color.

Informative outcome. Rook common-neighbor pairs are adjacent; Shrikhande common-neighbor pairs are not. Failure to reproduce that distinction or permutation invariance rejects the implementation, not a conjecture.

Bridge still needed. Unknown: identify what additional invariants and canonical choices suffice on all graphs while bounding total construction and branching time polynomially. A successful local invariant alone gives no such bridge.

Established / proposed. Color refinement and common-neighbor graph invariants are established. The card now labels the benchmark as reproduction, with any later signature extension requiring a literature audit.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

091 Unique Games Conjecture

Simulated probability and dynamics specialist

Target. For every $\varepsilon,\delta\in(0,1)$ with $\varepsilon+\delta<1$, does some alphabet size q make it NP-hard to distinguish unique-constraint instances with optimum at least $1-\varepsilon$ from those with optimum at most δ?

Restricted question. How do exact three-label optima and one fixed semidefinite relaxation differ on cycle versus complete base graphs with controlled edge permutations?

Approach. Define the relaxation and dataset before interpreting any gap. Use eight vertices and Gram variables $X_{(u,a),(v,b)}$. Require X positive semidefinite and entrywise nonnegative, orthogonal labels at each vertex, unit total diagonal per vertex, and equal vector sums across vertices. Maximize the normalized sum of constraint-compatible inner products.

Challenge. A relaxation gap is not computational hardness, and numerical solver output is not a certified upper bound. The conjecture's quantifiers cannot be inferred from one three-label dataset.

Next task. Enumerate all 3^8 assignments for exact optima. Solve the stated relaxation, enforcing equal vector sums through equal row-block sums; report numerical values separately from any certified dual bounds.

Baseline. Use C_8 and K_8, with identity permutations, powers of (012), or arbitrary S_3 permutations; four seed-91 instances per combination give 24 tests. Reverse-edge permutations are inverses.

Informative outcome. Any purported upper bound below the exact optimum exposes a formulation or certification error. Differences among the controlled families may motivate a precisely stated tractability lemma.

Bridge still needed. Unknown: prove hardness reductions with the required ε,δ and alphabet dependence, or construct algorithms for the entire promised class. Neither spectral observations nor finite integrality gaps provide that bridge.

Established / proposed. Wigderson describes unique games and semidefinite approximation. The finite structural comparison reproduces established methods and makes no claim of a new hardness result.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.math.ias.edu/files/Book-online-Aug0619.pdf

092 Quantum PCP conjecture

Simulated analyst

Target. For some fixed integer k and constants $0 \le a < b \le 1$, is it QMA-hard to distinguish $\lambda_{min}(H)/m \le a$ from $\lambda_{min}(H)/m \ge b$ for H on n qubits, given m=poly(n) k-local terms $0 \le h_i \le I$ with polynomial-bit encodings and $H = \Sigma h_i$?

Restricted question. How does one explicit positive-product amplification change normalized ground energy, locality, and term count for three specified four-qubit Hamiltonians?

Approach. Use $G_{i,j} = I - (I - h_i)(I - h_j)(I - h_i)$ for every ordered pair $i \ne j$, rather than leaving the gadget unspecified. Compare the new energy density with the old one. Verify positivity and the preservation of common zero vectors algebraically before diagonalization, while keeping spectral gaps separate from promise gaps.

Challenge. The construction can increase locality and term count without delivering a useful iterated reduction. An energy increase on solvable toy systems does not establish QMA-hardness.

Next task. Build exact rational matrices for A,B,C and their amplified sums. Compute interval-certified lowest eigenvalue bounds, count ordered-pair terms, and record each term's qubit support. Retain exact coefficient matrices.

Baseline. Let $P_i = |1\rangle\langle 1|_i$ and $Q_i = |-\rangle\langle -|_i$. Test $A = \Sigma P_i$, $B = \Sigma(P_i + Q_i)$, and $C = B + \Sigma R_{i,i+1}$, where R projects onto $(|00\rangle + |11\rangle)/\sqrt{2}$ and indices wrap modulo four.

Informative outcome. A negative eigenvalue outside numerical error or loss of a common zero vector rejects the implementation. Report energy-density gain jointly with support growth; either quantity alone is misleading.

Bridge still needed. Unknown: construct an efficiently computable, hardness-preserving map for arbitrary promised Hamiltonians whose normalized energy separation becomes constant while locality and encoding size remain appropriately bounded.

Established / proposed. The cited gap-amplification paper supplies established context. The explicit product audit is a toy calibration requiring its own proof and literature comparison.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2510.01333

093 Matrix multiplication exponent

Simulated experimental mathematician

Target. Is the matrix-multiplication exponent ω equal to 2: for every ε>0, can n×n matrices over Q be multiplied in O(n^(2+ε)) arithmetic operations?

Restricted question. Can a specified three-summand sub-tensor of the standard Strassen tensor square be replaced by two exact rank-one terms within a bounded coefficient alphabet?

Approach. Reclassify direct proportional-factor grouping as a negative control: standard Strassen factors are pairwise nonproportional in each mode, and so are the 49 factors in their tensor square. A richer experiment must replace whole sub-decompositions; an invertible common basis change or a symmetry relabeling cannot create literal proportionality.

Challenge. Failure of bounded three-to-two replacements is not a tensor-rank lower bound. Even a verified 48-term decomposition would improve one finite exponent bound, not prove ω=2.

Next task. Screen all three-term subsets by exact flattening ranks. For the first ten with every rank at most two, search two-term replacements using coefficients {-1,0,1} in fixed rational bases of their mode spans.

Baseline. The reviewer verified the seven-term identity and the 49 singleton factor classes. Preserve its exact ordered decomposition; do not rerun grouping as if successful merging were an open outcome.

Informative outcome. Reject any replacement failing coefficientwise tensor equality. A verified replacement lowers the total to at most 48; an empty screen or unsuccessful bounded search is recorded without extrapolation.

Bridge still needed. Unknown: obtain scalable decomposition families approaching exponent two. For the fixed experiment, first pin the source's term ordering, basis-normalization convention, and finite search grammar in code.

Established / proposed. Chiantini-Ikenmeyer-Landsberg-Ottaviani study Strassen decompositions and search geometry. The bounded replacement proposal carries no novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1610.08364

094 Exponential Time Hypothesis

Simulated combinatorics and complexity specialist

Target. Does the Exponential Time Hypothesis hold: no deterministic algorithm decides arbitrary n-variable 3-SAT in subexponential 2^o(n) time, apart from polynomial input-processing factors?

Restricted question. How do two fixed DPLL branching rules compare on a reproducible eight-variable sample, and do proposed clause-count potentials decrease on every observed branch?

Approach. Separate a concrete algorithm comparison from any asymptotic recurrence claim. Both solvers perform unit propagation in lexicographic order, branch false before true, and use no caching. Compare the least-index remaining variable with maximum clause-occurrence count, breaking ties by index; record the full branching tree.

Challenge. A sample of formulas cannot certify a worst-case recurrence. Faster average search or a smaller fixed exponential base neither establishes nor refutes ETH.

Next task. Compare decisions and recursive-call counts, including terminal calls. At every branch log $\mu=n+\alpha m_2$ for $\alpha\in\{0,1/4,1/2\}$, with m_2 counting current two-literal clauses after propagation.

Baseline. Generate 500 formulas with eight variables and twelve distinct three-literal clauses, each on distinct variables, using seed 94. Exhaustive truth tables provide satisfiability ground truth.

Informative outcome. A decision mismatch rejects the solver. Any observed increase rejects a claimed decreasing potential; surviving potentials require symbolic proof over all legal residual formulas before estimating an asymptotic base.

Bridge still needed. Unknown: prove a uniform worst-case algorithm with subexponential dependence on variables to refute ETH, or prove an unrestricted exponential lower bound to establish it. The sample only screens candidate analyses.

Established / proposed. DPLL, unit propagation, and measure-and-conquer are established methods. This controlled comparison is preparatory and makes no originality claim.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://www.scottaaronson.com/papers/pnp.pdf

095 Planted clique conjecture

Simulated probability and dynamics specialist

Target. For each fixed ε∈(0,1/2), is polynomial-time recovery of a uniformly planted clique of size floor(n^(1/2-ε)) in G(n,1/2) impossible with probability tending to one?

Restricted question. Can two explicit scalar statistics distinguish planted from null samples at the chosen finite sizes? This concerns detection; it does not recover the planted vertex set.

Approach. Define the statistics before generating data: E=Σi<j Bij and T=Σi<j<k BijBjkBki, where Bij=2 Aij-1 and Bii=0. Derive null expectations and use a separately generated training set to choose thresholds. Treat the study as an elementary detection baseline, not an unspecified corrected polynomial filter.

Challenge. Thresholded scalar statistics provide no recovery procedure. Finite detection performance may disappear asymptotically, and tuning on the test set would invalidate the stated comparison.

Next task. For every n,k use twenty null and twenty planted training graphs, then twenty each for testing, with separate seeds 95 and 1095. Choose training-optimal balanced-accuracy thresholds and freeze them.

Baseline. Use n∈{64,128,256}, k=floor(n^0.4) or floor(√n). Null graphs have independent fair edges; planted samples choose k vertices uniformly and force their internal edges.

Informative outcome. Report exact integer statistics, held-out confusion counts, and elapsed operations for each setting. Chance-like behavior rejects only this detector; a positive result still needs replication and asymptotic analysis.

Bridge still needed. Unknown: derive a recovery algorithm or an applicable detection-to-recovery reduction, then establish performance and polynomial complexity as n grows in the conjectured regime.

Established / proposed. The cited planted-structure paper develops established reductions. Edge and signed-triangle statistics are explicit baselines, not claimed new algorithms.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/1806.07508

096 Mandelbrot local-connectivity conjecture

Simulated probability and dynamics specialist

Target. For $M=\{c\in\mathbb{C}$: the orbit of 0 under $z\mapsto z^2+c$ is bounded$\}$, determine whether M is locally connected at every parameter $c\in M$.

Restricted question. Which geometric estimates in one bounded-type renormalization proof depend on a uniform bound on return times, and how do those dependencies enter parameter-space shrinking?

Approach. Replace an unspecified numerical example with a proof-dependency audit. Begin with an established bounded-type a priori-bounds result, trace its annulus estimates, and distinguish dynamical puzzle-piece shrinking from local connectivity in parameter space. Only after that audit should a new combinatorial restriction or recurrence be proposed.

Challenge. A finite list of return times cannot control all later scales. Dynamical-plane shrinking alone is not automatically the needed parameter-space conclusion, and a guessed recurrence may omit essential geometry.

Next task. Select the precise theorem and proof passage, enumerate every uniform constant and parameter-transfer step, and produce a dependency table before proposing a neighboring unbounded-type class.

Baseline. Dudko-Lyubich's bounded-type result, discussed in the cited survey, is the known theorem to reproduce. The survey's statement is context, not a substitute for its proof hypotheses.

Informative outcome. The deliverable is an exact location where a proposed extension loses a bound. If all steps remain within the known theorem, record reproduction rather than a new MLC case.

Bridge still needed. Unknown: supply geometric bounds for all remaining renormalization behavior and justify the parameter-space passage at every relevant parameter. An additional combinatorial assumption proves only a restricted result unless independently removed.

Established / proposed. Renormalization and a priori bounds are established. This revision commits to identifying the nearest theorem before suggesting an extension and makes no novelty claim.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2512.24171

097 Density of hyperbolicity for complex quadratic polynomials

Simulated analyst

Target. Are hyperbolic parameters dense in C for $z \mapsto z^2+c$, where hyperbolicity means the critical orbit converges to a finite attracting cycle or escapes to infinity?

Restricted question. Which points in three explicit rational grids near c=-3/4 admit a certified attracting fixed point or genuine two-cycle?

Approach. Use exact low-period algebra as the initial certificate engine rather than unspecified interval searches through period eight. Fixed points satisfy $z^2-z+c=0$; two-cycle candidates satisfy $z^2+z+c+1=0$. Check least period and multipliers strictly below one, recording boundary equality as unresolved rather than accepting it.

Challenge. Finite grids and low periods cannot establish density, and neutral parameters are expected to resist these certificates. Failure to find a cycle is not proof that a parameter is nonhyperbolic.

Next task. Compute algebraic fixed-point roots and multipliers 2z. For genuine two-cycles use multiplier 4(c+1); certify strict modulus inequalities by exact algebraic comparisons or enclosing rational intervals.

Baseline. Use c=-3/4+2^(-m)(a+ib), with m∈{4,6,8} and integers a,b∈{-2,-1,0,1,2}. The central parabolic parameter is an explicit nonacceptance control.

Informative outcome. Each accepted parameter receives a reproducible cycle certificate. False acceptance at the parabolic center rejects the implementation; unclassified points remain unclassified, not counterexamples to density.

Bridge still needed. Unknown: prove arbitrarily small hyperbolic perturbations near every complex parameter, including difficult recurrent behavior. These low-period certificates recover elementary neighborhoods only.

Established / proposed. Attracting-cycle multipliers and parabolic perturbations are established. This is a known-case certificate calibration, with no claim of a new density theorem.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/0805.1658

098 Furstenberg's ×2, ×3 measure conjecture

Simulated probability and dynamics specialist

Target. Every nonatomic Borel probability measure on R/Z invariant under T2(x)=2x and T3(x)=3x, and jointly ergodic for these maps, is conjectured to equal Lebesgue measure.

Restricted question. What can a relaxation using only Fourier frequencies of magnitude at most K distinguish, and what genuinely additional cross-scale information would its next inequality require?

Approach. Promote the reviewer's q-grid example to an analytic negative control. Uniform mass on q equally spaced points is invariant under doubling and tripling when gcd(q,6)=1, and its nonzero Fourier coefficients below q vanish. Therefore all constraints using that finite Fourier data agree exactly with Lebesgue measure.

Challenge. The grid is atomic and not jointly ergodic, so it is not a counterexample. Its indistinguishability invalidates atom-exclusion interpretations of finite data, not all possible cross-scale approaches.

Next task. Record the complete frequency inventory before any SDP. Design one explicit cross-scale inequality, stating every nonconcentration assumption and whether it follows from the conjecture's hypotheses; otherwise keep the task at protocol design.

Baseline. For K=12,24,48 use q=17,29,53. Moment matrices are indexed 0,...,K; impose invariance equalities only when every referenced index is ≤K, and use Fejér kernels with Fourier support in [-K,K].

Informative outcome. Predetermined: each stated grid survives every constraint that Lebesgue satisfies on this data. A claimed numerical separation indicates an extra frequency, an unstated assumption, or an implementation error.

Bridge still needed. Unknown: derive useful cross-scale control from invariance, nonatomicity, and joint ergodicity themselves, with a valid limiting argument. Arbitrary quantitative concentration assumptions can exclude precisely the hard cases.

Established / proposed. The Fourier formulation is established. The q-grid obstruction is an elementary review-derived control, not a novel theorem or a counterexample to Furstenberg.

Proposed route. protocol design; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2410.22701

099 Brennan's conjecture

Simulated analyst

Target. For every proper simply connected plane domain Ω and conformal bijection F:Ω→D, is ∫Ω|F′|^p dA finite for every 4/3<p<4?

Restricted question. For explicit reentrant-corner local models, do exact cutoff integrals reproduce the predicted upper-endpoint integrability threshold and its logarithmic boundary case?

Approach. Make the local map and normalization explicit: on the sector 0<arg z<απ, use f(z)=z^(1/α). This is a local corner model, not itself the global Riemann map onto D. Compute the integral over δ<|z|<1 exactly before considering nested fjord geometry.

Challenge. Local power models do not control constants across infinitely many corners or scales. They address the upper-endpoint mechanism, not the lower-endpoint issue for general unbounded domains.

Next task. Evaluate I=πα^(1-p)∫δ¹ r^[1+p(1/α-1)]dr symbolically, handling exponent -1 separately. Compare any numerical quadrature to this exact expression rather than inferring convergence from a plot. Normalize outputs by π and retain symbolic powers so rounding cannot determine the convergence classification.

Baseline. Take α∈{3/2,19/10,199/100,2}, p∈{7/2,39/10,4,41/10}, and δ=10^(-j), j=1,...,6. The α=2,p=4 case is a logarithmic-divergence control.

Informative outcome. Convergence requires 2+p(1/α-1)>0; equality gives logarithmic divergence. A mismatch rejects the quadrature or exponent calculation, while agreement only validates the known local model.

Bridge still needed. Unknown: prove global, summable distortion bounds for arbitrary domains, including interacting fjords and unbounded regions. An infinite-domain approximation would need uniform estimates before exchanging limits and integrals.

Established / proposed. Conformal corner asymptotics and Brennan's formulation are established. This explicit calculation is a calibration control, not a newly proved case of the full conjecture.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2604.12240

100 Hilbert's sixteenth problem, limit-cycle part

Simulated analyst

Target. For every degree bound d, does a finite H(d) bound the number of isolated periodic orbits of every planar real polynomial vector field of total degree at most d?

Restricted question. For $H=x^2/2+y^2/2-x^3/3$ and a fixed quadratic perturbation family, what does its first Melnikov function predict inside the period annulus $0<h<1/6$?

Approach. Specify $x'=Hy$ and $y'=-Hx+\varepsilon y(a+bx)$. With orientation given by the unperturbed flow, the first energy displacement is $M(h)=\oint_{H=h} y(a+bx)dx$. Derive this function and preserve the distinction between simple interior zeros, identically vanishing controls, and bifurcations near the center or separatrix.

Challenge. First-order zeros do not count all cycles, particularly at degeneracies or annulus endpoints. A local result for this quadratic family cannot bound every quadratic or higher-degree vector field.

Next task. Derive $M=2\int_{x^-}^{x^+} \sqrt{2h-x^2+2x^3/3}(a+bx)dx$ for the two roots bounding the center oval. Evaluate $h=j/600$, $j=1,\ldots,99$, with interval-certified quadrature.

Baseline. Use the five rational pairs (a,b)=(0,0),(1,0),(0,1),(-1/10,1),(-1/5,1). The zero pair gives M≡0; the (1,0) pair has positive M throughout the annulus.

Informative outcome. Check both controls exactly. Certified sign changes identify candidate interior zeros only; prove simplicity separately. Numerical uncertainty or M≡0 triggers higher-order analysis rather than an asserted cycle count.

Bridge still needed. Unknown: control all parameter degeneracies, both endpoints, and other polynomial systems uniformly in degree. Grid values do not supply that uniform bound or certify absence between samples.

Established / proposed. Melnikov analysis is established. This is a reproducible restricted benchmark, with no claimed new cyclicity bound or solution of Hilbert's problem.

Proposed route. calibration ready; readiness does not imply execution.

Completed result: none recorded.

https://arxiv.org/abs/2411.09594

Reproduce the work

Unpack the companion research pack into one folder. Numerical scripts require Python 3 and its standard library. Run each command as a separate line:

```
python3 verification_controls.py

python3 lang_trotter_control.py

python3 branching_support.py

python3 verify_branching_support.py

python3 composition_support.py
```

verification_controls.py runs the original finite checks and Collatz audit, compares whole JSON outputs to preserved baselines, and writes its named controls and environment manifest. The other scripts reproduce the new exact investigations; the independent support verifier must run after its enumerator.

The full pack includes all 100 revised records, the editable Markdown notebook, the 24 responses and item-level supplements, code and canonical outputs, retained baseline JSON, the new-pilot review and a SHA-256 file manifest. New script hashes and renderer dependencies are distinct from the numerical reproducibility manifest.

Optional PDF rebuild

Install ReportLab and pypdf. Bundled DejaVu fonts are included. Run python 3 build_revision.py; it writes the rebuilt PDF under output/. A PDF rebuild is separate from reproducing the mathematical outputs.

Execution environment and integrity

Python: 3.12.14 (main, Aug 25 2026, 14:00:49) [Clang 22.1.3] | Linux-6.18.35-x86_64-with-glibc 2.39

The initial paper and its peer review remain identifiable in the revision record. The peer-review document is unchanged. This edition incorporates criticism and completed finite investigations; closest-prior-work audits, unrestricted bridges and most proposed experiments remain outstanding.